The Capitalization of Local Energy Prices in the Housing Market

Felix Heuer[1,2]

[1]RWI - Leibniz-Institute for Economic Research
[2]Ruhr University Bochum

October 2025

Abstract

This paper investigates the capitalization of heating costs in the housing market. To identify the causal effect of heating costs, I use the gas price increase in 2022 and the war in Ukraine as a quasi-experimental setting. In a difference-in-differences (DiD) framework, I estimate the effect of rising gas-based heating costs on rental prices by comparing gas-heated apartments to those with district heating powered by renewable energy. To specifically analyze the impact on housing demand, I use time on market and number of clicks as alternative outcome variables. Local gas price data at the zip code level capture the local magnitude of the gas price shock and are used for estimating the treatment effect in a continuous DiD setting. The results reject the capitalization of heating costs into rental prices but show that housing demand is energy price elastic. Following the increase in gas prices, the time on market for gas-heated apartments rises by 12.5% on average. The effect varies with the size of the local gas price shock: an additional €100 in annual heating costs increases time on market by 0.6% on average and by up to 1% at the peak in April 2022.

1 Introduction

Despite significant growth in renewable energy over the past decade, residential heating is still predominantly dependent on fossil fuels. In the EU, for example, only 26% of heating and cooling is based on renewable energy sources in 2023 (Eurostat 2025*b*), while in the US, heating is also predominantly based on natural gas (51%) (EIA 2020).[1] Together, residential heating accounts for 25% of global energy consumption and 20% of global emissions in 2018 (International Energy Agency 2021). Achieving the climate goals of the 2015 Paris Agreement will therefore require the decarbonization of residential heating and enormous efforts over the next decade.

Against this background, this paper examines the capitalization of heating costs in the housing market and, in particular, whether housing demand responds to local energy prices. Understanding this relationship is essential for designing effective policies to support the transition to renewable heating sources. Price-based instruments are the most common strategy to encourage retrofits, improve energy efficiency, and reduce fossil fuel use.[2] However, these policies are only effective if energy costs are fully capitalized and households are aware of actual cost differences between heating sources (Myers 2019, Myers et al. 2022). This is especially important in rental markets, where landlords are likely to invest in energy-saving upgrades only if they can charge higher rent to cover the costs (Gillingham et al. 2012) — meaning tenants must be sensitive to heating costs and willing to pay more in exchange for lower energy bills (Lang & Lanz 2021). Given the urgency of decarbonizing residential heating, understanding how heating costs are capitalized is critical.

I estimate the effect of heating costs on rental prices (basic rent excl. heating costs)[3], as well as time on market and clicks per day as measures of housing demand, using a difference-in-difference (DiD) approach. To overcome the main challenge of identifying exogenous variation in energy prices, I use the Russian invasion in February 2022 as a quasi-experimental setting. The event was unexpected and led to an increase in the relative price of gas-based heating compared to district heating, as shown in Figure 1. Comparing gas-heated apartments with renewable district heating (DH) apartments, therefore, allows to identify the causal effect of energy prices in a DiD framework. Renewable DH is defined by the fossil fuel share using a novel dataset containing information on 1001 individual DH networks. Networks with a fossil fuel share of at most 30% are defined as renewable. Together, gas-heated (56%) and DH (15%) apartments make up the largest share of apartments in Germany and represent the largest groups of fossil and renewable heating sources, respectively.[4] To ensure substitutability and equal local availability between the two apartment types, I focus only on neighborhoods where both renewable DH and gas heating are available. I further address remaining structural differences between gas-heated and district heating apartments by using an entropy balancing approach to strengthen the causal claim of the analysis.

In addition, I exploit spatial differences in the magnitude of the gas price shock to estimate the capitalization of local energy prices. To this end, gas price data at the zip code level are obtained from the online platform *Verivox*. Real estate data for the period from 2021 to 2022 are taken from the *RWI-GEO-RED* dataset, which includes all listings from *Immoscout24.de* (RWI – Leibniz Institute for Economic Research & ImmobilienScout24 2024), the German market leader in housing advertisements. The main analysis of this paper is based on rental apartments.[5]

My results reject the capitalization of heating cost in rental prices and show only a small insignificant

[1] The second largest energy source for residential heating in the US is electricity, which is predominantly generated from natural gas (40%) and with only about 20% coming from renewable energy sources (EIA 2020).

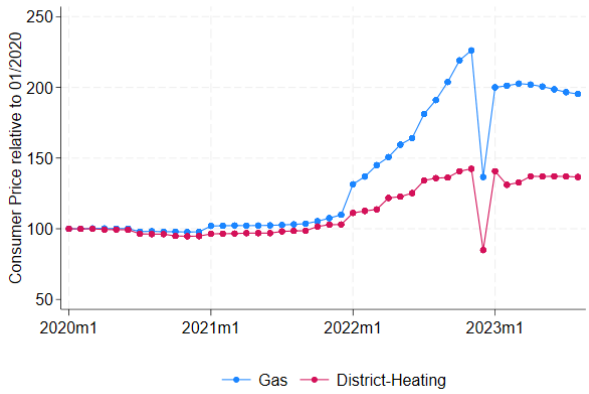
[2] Despite some attention of policy maker to ban fossil-fuel based heating systems, a broad resistance prevents such interventions. For a more detailed discussion in the costs and benefits of bans relative to price-based policies refer to Edenhofer et al. (2025). For a more detailed assessment of climate policies refer to Stechemesser et al. (2024)

[3] Throughout this paper, rental prices refer to the basic rent excluding heating costs and other ancillary costs.

[4] Based on 2022 German census data.

[5] Results for the sales market for both apartments and properties are available upon request, but show no significant effect of heating costs on any of the three outcomes (listing price, time on market, and clicks).

Figure 1: Avg. Consumer Prices from the Federal Statistical Office of Germany.



negative effect of heating costs on rental prices after February 2022. Instead, I find an increase in the time on market of gas-heated versus renewable DH apartments due to an increase in gas-based heating costs of 12.5% on average and 19.8% at the peak in August 2022. The effect on clicks is close to zero and statistically insignificant. The results thus suggest that despite the energy price elasticity of housing demand, rental prices do not reflect relative heating cost differences.

With respect to local energy prices, an additional €100 increase in heating costs increases time on market by 0.6% on average. I find no differences in rental prices or clicks due to differences in the magnitude of local gas price shocks. This supports the finding that housing demand adjusts to changes in local heating costs, while rental prices do not reflect differences in energy prices.

This paper contributes to several strands of the literature. First, it primarily adds to previous findings on the capitalization of heating costs in property prices and the valuation of heating costs. Myers (2019) uses year-by-year changes in the relative price of oil and gas in Massachusetts, US, to estimate the effect of energy costs on property prices. The estimates show a significant decrease in sales prices of gas-heated properties in years with a relatively high gas prices and full capitalization of heating costs in property values. Similarly, a large number of studies find that home buyers are willing to pay a price premium for energy-efficient homes[6] or homes with cheaper or renewable heating systems[7], which is often interpreted as a willingness to pay for lower heating costs and insurance against future energy price shocks (Eichholtz et al. 2013, Fuerst et al. 2015, Kholodilin et al. 2017). For example, Harjunen & Liski (2014) find a price premium for district heating that is near the capitalized value-difference in heating costs in the Finish context. Taken together, these studies suggest that households and housing demand do take heating costs and energy prices into account. However, these studies do not rely on local energy prices, which may be less salient to households and therefore may result in lower capitalization rates.

With respect to the capitalization of local heating costs, Mense (2018) suggests that local energy prices and local weather conditions are not taken into account in the valuation of energy efficiency - being the only study to my knowledge leveraging local energy prices. Using gas price data at the zip code level for Germany between April 2015 and July 2016, Mense (2018) estimates the effect of changes in local gas prices over time on listed property prices by energy efficiency scores for gas-heated apartments. On the one hand, the results support evidence that home buyers, on average, are willing to pay a premium for

[6]See for example, Brounen & Kok (2011), Deng et al. (2012), Zheng et al. (2012), Eichholtz et al. (2013), Kahn & Kok (2014), Fuerst et al. (2015), Kholodilin et al. (2017), Fuerst & Warren-Myers (2018), Mense (2018), Aydin et al. (2020), Myers et al. (2022), Taruttis & Weber (2022), März et al. (2022), Sieger & Weber (2023), Amaral et al. (2024), Sejas-Portillo et al. (2025).

[7]See for example, Harjunen & Liski (2014), Shen et al. (2021), Gillingham & Watten (2024), or Ruokamo & Sahari (2025).

energy efficiency. On the other hand, they do not seem to take local conditions into account and energy efficiency is not valued more in locations with higher gas prices or colder weather conditions.

This paper contributes threefold to this strand of the literature. First, this paper focuses on the rental market, which is relatively understudied but accounts for a large share of the total population, especially in Germany (52.3%) and other European countries (Eurostat 2025*a*).[8] Insights from the rental market are particularly important in the context of distributional effects across income groups, as low-income households are more likely to rent and are relatively more affected by changes in heating costs (Campagnolo & De Cian 2022, Guan et al. 2023, Zhu et al. 2024). Second, I focus on housing demand and the effect of local energy prices not only on rental prices but also on time on market and listing clicks. This allows to address the role of search frictions, where energy prices may influence not only equilibrium prices but also the speed of transactions — that is, time on market — as landlords may be willing to trade higher rents for longer search durations (Haurin 1988, Genesove & Han 2012). Alternatively, changes in equilibrium prices might underestimate the demand effect. For example, Carroll et al. (2024) show that energy-efficient properties in Ireland between 2017 and 2018 were not only sold at a premium but also spent less time on the market. Third, combining new data on the local availability of renewable DH with local gas price data allows to identify the causal effect of energy prices in a quasi-experimental setting. Thereby, the Russian invasion in February 2022 and the associated gas price increase serves as treatment with varying treatment intensity across zip code areas. Identification relies on temporal changes in relative heating costs by fuel type within zip code areas.[9]

Another broad literature this paper contributes to examines the elasticity of residential gas demand. Evidence consistently shows that households are price sensitive and reduce their gas consumption in response to higher prices.[10] Beyond consumption reductions, Ahlvik et al. (2025) show that households adjust along multiple margins when faced with higher energy costs. Using the 2022 European Energy Crisis and the associated spike in electricity prices, they identify four such margins in the Finnish context: energy use, earnings, financial distress, and residual consumption. Similarly, Sahari (2019) and Carroll et al. (2024) demonstrate that rising energy costs can also lead to increased investment in low-energy-use technologies, pointing to a multifaceted behavioral response to heating cost shocks. This paper extends this literature by examining a distinct adjustment margin: the choice of heating source, specifically among households in the process of searching for a (rental) home. By focusing on this margin, the paper sheds light on how energy price signals affect longer-term, structural decisions in the housing market.

Lastly, this paper contributes more broadly to the literature on tax incidence[11] and the effects of price-based policies on the housing market (Kryzanowski et al. 2024, Kholodilin 2025). While focusing on short-term effects, the findings of this paper suggest that a policy-induced increase in fossil fuel prices — such as through carbon pricing — is not capitalized into rental prices. As a result, higher energy costs do not appear to incentivize landlords to invest in new heating systems or energy-efficiency

[8]For the rental market, several studies estimate the willingness to pay for energy efficient homes (e.g., März et al. 2022, Sieger & Weber 2023) or refer to the so-called "energy efficiency gap" (Allcott & Greenstone 2012, Ramos et al. 2015, Lang & Lanz 2021, Bishop & Kiribrahim-Sarikaya 2024, Gerarden et al. 2017, Galvin 2023) and the "landlord-tenant-dilemma" (Gillingham et al. 2012, Myers 2020, Myers et al. 2022, Singhal et al. 2025). These studies argue that asymmetric information and a split-incentive problem prevent landlords from making investments in energy efficiency improvements whose benefits accrue to tenants in the form of lower heating costs (Ramos et al. 2015, Myers 2020, Lang & Lanz 2021, Davis 2023, Lambin et al. 2023). My results add to this literature (i) by showing that an inelastic demand for housing with respect to heating costs does not explain the lack of investment in rental housing and (ii) being the first study to estimate the effect of heating costs on rental prices.

[9]Studies on the price premium for district heating generally do not take into account differences in local energy prices (Harjunen & Liski 2014, Yoon et al. 2015, Krikser et al. 2020). Similarly, Braakmann et al. (2023) use the Russian invasion as a quasi-experiment to estimate the effect of increased heating costs on the demand for energy-efficient homes in the UK, but do not exploit local price variations.

[10]See, for example, Alberini et al. (2020), Burns (2021), Favero & Grossi (2023), Rubin & Auffhammer (2024), Jamissen et al. (2024), Behr et al. (2025), and Ahlvik et al. (2025).

[11]See for example, Kopczuk & Munroe (2015), Slemrod et al. (2017), Best & Kleven (2018), Fritzsche & Vandrei (2019), Löffler & Siegloch (2021)Millar & White (2024), Lyu (2024), or Dolls et al. (2025) among others.

improvements. Instead, these costs are largely passed on to tenants in the form of higher heating bills. The absence of heating cost capitalization in rents raises concerns about the effectiveness of price-based policies in promoting energy efficiency in the rental housing sector. Moreover, with absent heating cost capitalization of heating changes in rental prices but an elastic demand, the shift away from relative more expensive (fossil) fuels will only be possible for some households. Likely, leaving disadvantaged and low-income households households exposed to carbon pricing and higher fossil prices.

The remainder of the paper is organized as follows. Section 2 describes a search and matching model to motivate the effect of local energy prices on the rental market in a partial equilibrium setting. Section 3 first describes the data used and the empirical strategy, followed by the corresponding estimation results. Section 4 discusses the estimation result in the context of the search and matching model and section 5 concludes.

2 Partial Equilibrium Model

In this section, I describe a partial equilibrium model of the rental market to predict the effect of rising heating costs and the effect of spatial differences in local heating costs on the rental market. The model is based on a random search and matching model (Diamond 1982, Mortensen 1982, Pissarides 1985, Mortensen & Pissarides 1994), which has been applied to the housing market by Wheaton (1990), Krainer (2001), Piazzesi & Schneider (2009), or Genesove & Han (2012), among others. The model in this paper includes additional features to capture the unique characteristics of the rental market, namely a non-negotiable rental price and period-wise rental payments, as well as a differentiation of total rental payment in rental price and utility costs.

In the model T tenants search for rental apartments offered by L landlords across different neighborhoods R and market segments N. Thus, the rental markets is divided in different submarkets M differentiated by locations and apartments types. Both groups of agents are risk neutral. The number of agents and their distribution is exogenous. Heating costs h_{rs} are determined by neighborhood specific energy prices and the heating type s which determines the reliance on energy prices, e.g., capturing differences between fossil and non-fossil heating systems.[12]

TENANTS

Tenants decide in which neighborhoods to search for housing based on their location preferences. Tenants may search in multiple neighborhoods, but not all tenants search in all neighborhoods. This location decision is assumed to be exogenous and independent of landlord behavior. For example, consider a setting where tenants are only willing to live in certain neighborhoods due to commuting constraints or social ties.

Similarly, tenants restrict their search to market segments based on their household characteristics, such as household size or income level. The housing market can therefore be divided into $M = R \times N$ submarkets across neighborhoods and market segments, following the notion of segmented housing search of Piazzesi et al. (2020). In the context of housing search on online platforms such as *Immoscout24.de*, tenants can select the submarket directly through search filters based on some key characteristics (e.g. location, type of apartment, number of rooms, etc.). Tenants select some search criteria and consider only apartments with some pre-defined characteristics, i.e., apartments in certain submarkets.

Each tenant i has an idiosyncratic valuation v_{ij} for an apartment j based on its attributes $f \in F$. These idiosyncratic valuations are drawn from the distribution $g_{V|F}(v|f)$ and describe i's willingness

[12] Alternatively, apartment types might also capture differences in energy efficiency instead of heating systems as in Myers (2020).

to pay for apartment j with attributes f.[13] Relative to the overall distribution of v across all sets of characteristics F, $g(v)$, the conditional distribution $g_{V|F}(v|f)$ is restricted to a submarket specific interval between zero and $\bar{v}_m$. In other words, conditioning on $F = f$ defines the maximum willingness to pay among the subset of tenants searching in submarket m, that is, in neighborhood r and market segment n. $\bar{v}_m$ represents the maximum valuation among all tenants searching in a submarket m. This reflects differences in the valuation of apartment characteristics within and across different submarkets of the housing market.[14] The conditional distribution for each submarket is known to both tenants and landlords.

The tenant's net willingness to pay for an apartment is adjusted for heating costs and is given by $x_{ijr} = v_{ij} - \kappa h_{rs}$. Heating costs h_{rs} are exogenous, neighborhood and heating type specific, and determined by local energy prices as well as the cost effectiveness of the heating type. The perception parameter κ captures potential distortions in tenants perception of (future) heating cost, e.g. taking into account imperfect information, limited salience, or other behavioral biases. Tenants underestimate future heating costs for $\kappa < 1$ and overestimate actual costs for $\kappa > 1$. For simplicity, energy consumption is assumed to be homogeneous across tenants and independent of apartment characteristics f, such that v and h are independent. Idiosyncratic valuations v only captures tenants' willingness to pay for an apartment irrespectively of heating costs which are valued independently.

LANDLORDS

Each landlord j offers one apartment with heating type s in submarket m (neighborhood r and market segment n), specifying the rental price excluding heating costs, p_j, and a set of apartment-specific attributes f.[15] Misreporting and selective reporting of attributes are assumed to be strictly unbeneficial. The assumption of truthful reporting allows simplifying the search process from a two-step game, with initial contact and viewing, to a one-step game, where tenants receive all relevant information ex ante. The rental price is non-negotiable and is set ex ante under full information about the number of tenants searching in m, T_m, the number of competing landlords L_m, and the conditional distribution of valuations $g_{V|F}(v|f)$ given heating costs h_{sn}. Asymmetric information about local heating costs or tenants' expectations of future heating costs can lead landlords to misjudge the demand elasticity with respect to heating costs, causing their perceived heating-cost elasticity $\hat{\kappa}$ to differ from the true demand elasticity of tenants κ. Put differently, landlords set rental prices based in their anticipation of tenants heating cost elasticity under incomplete information.

EQUILIBRIUM CONDITIONS

Equilibrium prices and allocations are determined for each submarket m similar to the process of a random search and matching model (Diamond 1982, Mortensen 1982, Pissarides 1985, Mortensen & Pissarides 1994). The housing market tightness in submarket m is given by $\theta_m = \frac{T_m}{L_m}$. Let $\vartheta(T_m, L_m)$ denote the rate of contacts between tenants and landlords in submarket m. The function $\vartheta()$ is assumed to exhibit constant returns to scale and to increase in both arguments. However, not all contacts lead to a transaction, depending on the relation between the listed rental price and the tenant's valuation of apartment j.

The probability that landlord j is contacted by a tenant is given by the following matching probability:

$$q_m(\theta_m) = \frac{\vartheta(T_m, L_m)}{L_m} = \vartheta(\theta_m, 1) \tag{1}$$

[13] The apartments characteristics are purely aesthetic in nature.
[14] Williams & Capital (2014) use the truncation of the distribution under the notion of focused search.
[15] Subscript j refers to both landlords and apartments, since each landlord offers only one apartment.

with $q'_m(\theta_m) > 0$ since $\vartheta()$ is increasing in both arguments. Thus, a decrease in the housing market tightness, or fewer tenants per landlord in m, decreases the probability that a landlord will be contacted by a tenant, i.e., the matching probability.

For a given match of landlord and tenant, a transaction occurs if and only if the listed rental price is less than or equal the tenant's heating-cost-adjusted willingness to pay. This is in contrast to the common search and matching model where prices are subject to a bargaining process. While this may be appropriate for the sales market, the rental market is best represented by a take-it-or-leave-it offer.

The probability of a transaction conditional on a match can be derived as follows, under the assumption that tenants' willingness to pay, v, is independent and identically distributed:

$$\begin{aligned} Pr(transaction) =& Pr\left(p_j \leq x_{ij}|f\right) \\ =& 1 - Pr\left(v_{ij} \leq p_j + \kappa h_{rs}|f\right) \\ =& 1 - G_{V|F}\left(p_j + \kappa h_{rs}|f\right) \end{aligned} \tag{2}$$

where $G_{V|F}(p_j + \kappa h_{rs}|f)$ is the conditional cumulative distribution function (CDF) of $g_{V|F}(v|f)$. The transaction probability in each period is therefore a product of the matching probability and the conditional transaction probability: $q_m(\theta_m)\left(1 - G_{V|F}\left(p_j + \kappa h_{rs}|f\right)\right)$. As expected, the transaction probability decreases with a higher rental price or expected heating cost, as well as with a lower housing market tightness in m.[16] Figure A2 shows the cumulative transaction probability under the distribution assumption $v \sim U(0, \bar{v}_m)$. First, given a lower housing market tightness and fewer tenants per landlord, landlords are less likely to be contacted, which prolongs search and time on market. Second, for a given matching probability, the transaction probability decreases with the rental price or heating cost because fewer tenants are willing to rent at a given price. An increase in κ has the same effect. In all three cases, the transaction probability decreases in each period so that the cumulative probability converges to one at a lower rate, increasing search duration measured in time on market.

In case of no transaction, with probability $1 - q_m(\theta_m) + q_m(\theta_m)G(p_j)$, the game repeats: the landlord is matched with another tenant with probability $q_m(\theta_m)$ which leads to a transaction with probability $1 - G_{V|F}(p_j + \kappa h_{rs}|f)$. The landlord, therefore, faces a trade off between raising the rental price to maximize rental payments against keeping the rental price low to increase the transaction probability and to obtain rental income sooner.

The maximization problem of a landlord is defined with $G(p_j) = G_{V|F}\left(p_j + \kappa h_{rs}|f\right)$, $q_m = q_m(\theta_m)$, and discount rate $\beta \in (0, 1)$ as follows:[17]

$$\begin{aligned} \max_{p_j} U(p_j) &= \overbrace{q_m\Big(1 - G(p_j)\Big)\sum_{k=0}^{\infty}\beta^k p_j + \ldots}^{t=0} + \overbrace{\Big(q_m G(p_j) + 1 - q_m\Big)^n q_m\Big(1 - G(p_j)\Big)\sum_{k=n}^{\infty}\beta^k p_j + \ldots}^{t=n} \\ &= p_j \frac{q_m\Big(1 - G(p_j)\Big)}{1 - \beta - (\beta - \beta^2)(q_m G(p_j) + 1 - q_m)} \end{aligned} \tag{3}$$

landlord j sets the rental price p given the matching probability q_m to increase future discounted rental income $\sum_{k=0}^{\infty}\beta^k p_j$ but limiting a the associated decrease in the transaction probability $1 - G(p)$. Figure A3 displays equation (3) for the distributional assumption $v \sim U(0, \bar{v}_m)$. Over the feasible set of prices $p \in [0, \bar{v}_m - h_{rs}]$, landlords' expected lifetime utility $U(p)$ is a concave function of the rental price, with p^* being the utility maximizing rental price. For $p \in [0, p^*)$, an increase in the rental price increases rental payments and expected utility. For $p \in [p^*, \bar{v}_m - h_{rs}]$ the direct effect of higher rental payments is dominated by a reduction in the transaction probability and utility decreases with a higher rental price.

[16] Notice, heating cost induced changes in tenant behavior might stem from actual or perceived cost, if $\kappa \neq 1$.

[17] A more detailed derivation of the maximization problem is given in Appendix A.1. In Appendix A.2, the model is extended by allowing for limited contract duration using a Calvo-type price setting.

For $p > \bar{v}_m - h_{rs}$ no tenant searching in m is willing to rent and both the transaction probability and expected utility are zero.

An increase in (perceived) local heating costs reduces the maximum rental price for which there exists a positive transaction probability and therefore reduces expected lifetime utility of a landlord. A decrease in $\bar{v}_m$ has the same effect. A reduction in the discount rate β reduces the valuation of future income and increases the cost of a delayed transaction such that $U(p_j)$ decreases in β. Similarly, a reduction in the matching probability, q_m, due to a lower housing market tightness also reduces landlords lifetime utility as it becomes more difficult to be matched with a tenant.

Taking the first derivative of equation (3) with respect to p_j yields the optimal rental price, p*, which maximizes $U(p_j)$. For the distributive properties of $v \sim U(0, \bar{v}_m)$ with $G(p_j) < 1$ and $\beta < 1$ the equilibrium rental price is given by:

$$\begin{aligned} p_j^* &= (\bar{v}_m - \hat{\kappa} h_{rs}) \frac{1 - \sqrt{1 - \beta q_m (1 - \beta(1 - q_m))}}{\beta q_m} \\ &= \bar{x}_m \Phi_m \end{aligned} \tag{4}$$

The optimal rental price increases linearly in the maximum willingness to pay among tenants $\bar{v}_m$ and decreases linearly in (perceived) local heating costs κh_{rs}. In both cases, an increase in $\hat{\kappa} h_{rs}$ or a decreases in $\bar{v}_m$ reduces the maximum rental price for which landlords face a positive transaction probability which limits rental income potential and reduces transaction probability for a given price. Notice that $\hat{\kappa}$ refers to the landlord's expectation on tenants' heating cost elasticity. For simplicity assume for know that $\hat{\kappa} = \kappa$.

With respect to the discount rate β and the matching probability q_m, the optimal rental price increases in both arguments (Figure A4). First, an increase in β increases the valuation of future rental income, makes a delayed transaction less costly, and reduces the negative impact of a lower transaction probability. Similarly, the optimal price increases in q_m as transactions for a given price become more likely in all periods.

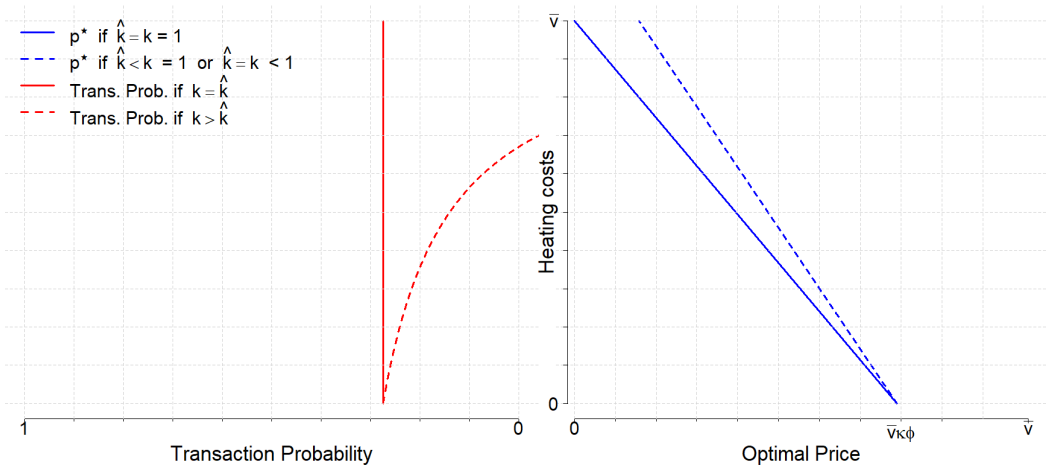
Compared to classic search and matching models, Φ_m denotes the total surplus share earned by landlords. For q_m and β equal to one, landlords extract the maximum rental price $p^* = \bar{x}_m$. Since tenants are assumed to be indifferent between heating costs and rent, Φ_m also represents the share of heating costs borne by landlords, depending on the heating cost elasticity κ. In other words, a one-unit increase in heating costs lowers the equilibrium rental price by $\kappa \Phi_m$ but tenants utility by κ.

MODEL PREDICTIONS

The model's central prediction is that the optimal rental price p^* is a fixed fraction Φ_m of the maximum heating-cost-adjusted willingness to pay, $\bar{x}_m$. Landlords do not list the maximum feasible rental price; instead, they post a markdown to shorten search duration. Thereby they account for their intertemporal discount rate and the matching probability, which together capture their bargaining power. Assuming exogenous market tightness in m, this markdown is independent of heating costs, so the rental price adjustment is smaller than the change in heating costs. Consequently, tenants bear the share $1 - \kappa \Phi_m$ of an increase in heating costs and the equilibrium rental price decreases proportional to a change in heating costs.

Figure 2 depicts the equilibrium relationship between heating costs, transaction probability, and rental price. It contrasts the case where landlords' expected heating-cost elasticity of demand matches the true elasticity ($\kappa = \hat{\kappa}$) with the case where landlords underestimate tenants' heating costs elasticity ($\hat{\kappa} < \kappa$). In both scenarios, higher heating costs reduce the rental price, but the reduction is smaller when $\hat{\kappa} < \kappa$, because landlords lower prices only in line with their expected demand response. Only if

Figure 2: Predicted Relationship between Heating Costs, Rental Price, and Time on Market



housing demand or supply is fully inelastic to local heating costs—i.e., when $\kappa = 0$ or $\hat{\kappa} = 0$—does an increase in heating costs leave the equilibrium rental price unchanged.

When landlords' expectation of tenants' heating-cost elasticity is accurate ($\hat{\kappa} = \kappa$), the lower rental price offsets the heating-cost-induced decline in the pool of tenants willing or able to rent. This keeps the transaction probability constant and preserves the optimal balance between price and search duration. This is because p^* remains the same fraction of the maximum feasible rental price, and therefore the share of tenants willing or able to rent stays constant. In other words, the probability that a matched tenant accepts the rental price is unchanged. Heating costs reduce overall welfare, but the landlords' share remains constant, as they do not alter tenants' intertemporal trade-off between rental price and search duration.

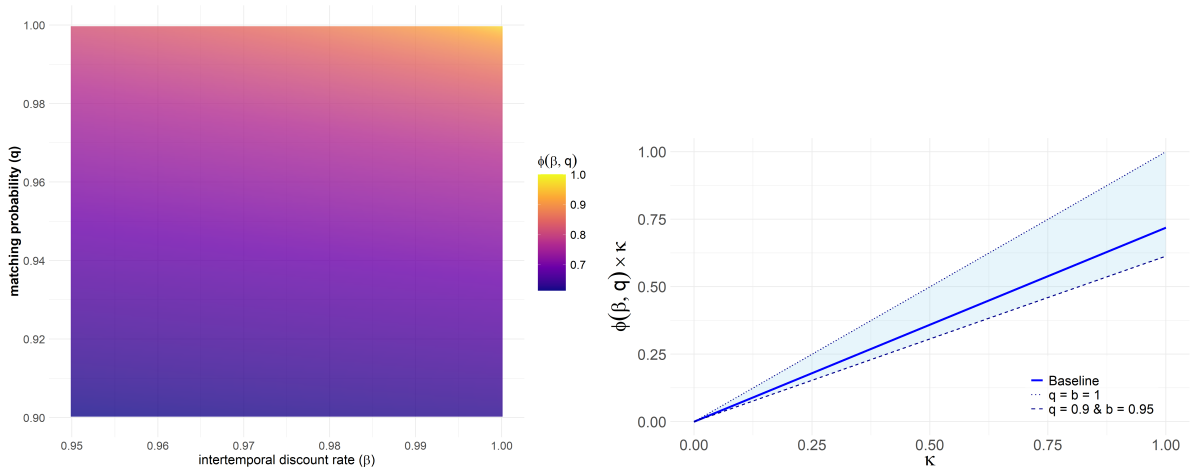
If landlords misjudge tenants' heating-cost elasticity ($\hat{\kappa} < \kappa$), the price reduction is insufficient, shifting a larger share of heating costs to tenants, which reduces the transaction probability and prolongs search duration. A smaller share of tenants than expected is willing or able to accept the rental price. Divergent expectations about future heating costs may arise if landlords are less informed about local conditions. Possible reasons include living elsewhere, not paying utility bills directly, or facing weak incentives to acquire costly information (Sallee 2014, Angeletos & Sastry 2025). -As a result, landlords may respond less to changes in local heating costs than tenants, underestimating demand shifts and adjusting rental prices inadequately. Alternatively, landlords may overestimate tenants willingness of acceptance of saving energy in response to rising energy prices.

Overall, the model predicts a negative relationship between heating costs and rental prices, and a negative effect on transaction probability (time on market) only when landlords misjudge tenants' heating-cost elasticity.

To gauge the magnitude of predicted rental price changes, Figure 3 reports, for realistic parameter values, the markdown Φ and the heating cost pass-through rate $\Phi\hat{\kappa}$. The markdown at which rental prices are set below the maximum feasible rental price directly determines how changes in it influence rental prices. For matching probabilities (q) between 90 and 100% and intertemporal discount rates (β) between 0.95 and 1, the optimal rental price is set at a markdown of 0.62 to 1 relative to the maximum feasible price (Figure 3(a)). Relative to β does q have the larger impact on Φ—and therefore on the rental price- causing a wide range of potential price effects. For example, with a common discount rate of 0.95, landlords set rental prices at a markdown between 0.62 and 0.87, depending on the sub-market-specific matching probability.

Next, Figure 3(b) plots the pass-through of heating cost changes on rental prices for $\hat{\kappa} = \kappa$ and

Figure 3: Predicted Magnitude of Markdown and Heating-Cost-Pass-through



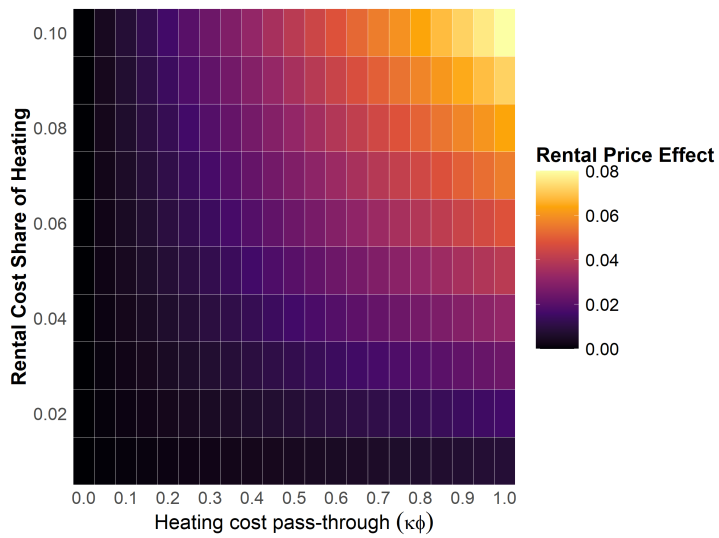
(a) Predicted Markdown (Φ)

(b) Predicted Heating-Cost Pass-through ($\Phi\kappa$)

different levels of tenants' heating cost elasticity and markup scenarios. When landlords set the maximum feasible rental price ($q = \beta = 1$), the rental price changes by exactly κ after a one-unit increase in heating costs, because higher heating costs reduce the maximum feasible rental price by this factor. In scenarios with a markdown ($\Phi < 1$)—that is, when $q < 1$ and $\beta < 1$—the rental price changes by more than κ, reflecting that some of the heating cost increase is borne by the landlord. For the baseline specification with $\beta = 0.98$ and $q = 0.95$ and heating cost elasticity of close to one, heating cost are passed onto renal prices by a factor of about 0.72.

To predict the impact of the 2022 gas price increase on rental prices, we begin with the fact that heating costs account for roughly 9% of total rental costs (warm rent including all ancillary charges). An 80% rise in gas prices therefore raises total rental costs by about 7.2% on average. Within the model, this translates into a $\hat{\kappa} = 7.2\%$ reduction in the maximum feasible rental price. Assuming a pass-through rate of about 0.72, we consequently expect the rental price for gas-heated apartments to decline by roughly 5.2% relative to apartments unaffected by gas prices.

Figure 4: Predicted Rental Price Effect of 2022 Gas Price Increase



Notes: The predicted rental price effect is based on a 80% increase gas price increase during the 2022 increase in consumer gas prices.

Figure 4 shows the predicted price effect for alternative heating-cost shares of total rental costs and different pass-through rates. The largest rental-price decrease of around 8% occurs under full pass-through when heating costs represent 10% of total rental expenses. By contrast, when the pass-through rate is below 0.1, or when heating costs make up less than 1% of total rental expenses, the predicted effect is effectively zero.

3 Estimating the Effect of Energy Price

In this section, I discuss my empirical strategy for estimating the causal effect of heating costs on rental prices, time on market, and clicks using a change in gas-based heating costs. Afterwards, I present the corresponding estimates of the effect of heating costs.

3.1 Data

3.1.1 Real Estate Data

The analysis of this paper is based on the *RWI-GEO-RED* dataset (RWI – Leibniz Institute for Economic Research & ImmobilienScout24 2024), which covers all apartments listed for rent on *Immoscout24.de*.[18] For the observation period from January 2020 to November 2022, the dataset includes a total of 2.3 million apartments listed for rent and provides a good representation of the online rental market for apartments in Germany.[19,20] In light of the research question at hand, the data provide three key insights into the housing market. First, the data include detailed information on the built-in heating system and the type of fuel used. This makes it possible to differentiate apartments by heating type and their exposure to gas price changes. Secondly, the rents are given with and without ancillary costs, which makes it possible to estimate the effect of energy prices on the basic rent, independent of increases in heating costs or other ancillary costs.[21] In the following rental price is used as a synonym for basic rent. Third, the data include information on the time on market of an apartment and the number of clicks a listing generates per day. Both are measures of housing demand. In the context of the model described in section 2, time on market serves as a proxy for search duration, or transaction probability. Clicks represent a measure of tenants submarket choice since the number of clicks is mostly determined by some key characteristics and the search filters available on the *Immoscout24* website. Additional apartment-specific characteristics in the data include floor space, number of rooms, year of construction and last modernization, facilities and renovation level, etc. The geo-coordinates of the apartments are used to control for regional characteristics at the one square kilometer grid level based on the *RWI-GEO-GRID* dataset (Breidenbach & Eilers 2018) and to link them to the local availability of renewable district heating and local gas prices.

The dataset is cleaned in the following way. First, observations with missing values for key variables are removed, namely observations without location information (longitude or latitude), floor space, rental price, start or end date, or reported clicks per day. Second, outliers or observations with unusual characteristics are dropped. These are apartments in buildings built before 1500, the top and bottom one percent in terms of rental price and floor space, and the top one percent in terms of listing duration and clicks per day. Third, duplicates are removed based on the start and end dates of a listing.

[18] For a more detailed description on the data set refer to Thiel (2024).

[19] Months after November 2022 are excluded for two reasons to avoid biased estimates. First, due to the introduction of different policies to subsidize private gas consumption and limit heating cost increases. Second, Figure A1 shows a decline in the number of gas-heated apartments listed after December 2022, suggesting misreporting of heating types.

[20] The market share of Immoscout24 is over 70% according to the latest data from the German Federal Cartel Office (Federal Cartel Office 2016).

[21] Observations are dropped (18% of the sample) if there is a risk that heating costs are included in the basic rent.

3.1.2 Energy Price Data

Data on energy prices are obtained from three different sources. First, national average consumer prices for gas and district heating are obtained from the German Federal Statistical Office. Second, local gas price data are taken from the online platform *Verivox*. The data contain gas price data of the main supplier at the zip code level and represent the annual heating costs for a household with an annual consumption of 20,000 kWh.[22] Third, district heating data are collected from the German Energy Efficiency Association for Heating, Cooling and Cogeneration (AGFW). The dataset contains information on 1001 district heating networks in 425 municipalities, including several network-specific characteristics such as energy source and share of renewables, price (ct/KWh), network size (MW), etc. The network level data has been geocoded at the municipality level which allows to identify the local fuel mix of district heating and the existence of renewable DH.

3.2 Empirical Strategy

3.2.1 Treatment and Control Group

I use a difference-in-difference (DiD) strategy to estimate the impact of heating costs on different housing market outcomes. Specifically, I compare gas-heated and renewable DH apartments using the gas price increase associated with the Russian invasion in Ukraine in February 2022 as the treatment. The increase of gas-based heating costs relative to renewable district heating allows to estimate the effect of heating costs on rental prices, as well as time on market and clicks as measures of housing demand.

The identification strategy is based on the idea of finding two comparable groups of apartments, only one of which experiences an energy price-related increase in heating costs. Gas-heated and renewable DH apartments are used for the treatment and control groups, respectively, for two reasons. First, both gas-heated (56%) and district-heated (15%) apartments represent the largest share of apartments in Germany and are the largest groups of fossil and renewable heating, respectively. Second, only the heating costs of gas-heated apartments are directly affected by the gas price shock in 2022. In contrast, district heating prices based on renewable energy sources are not affected by gas prices. The pricing of district heating is generally based on a network and supplier-specific pricing function that includes a base price for operating costs (including infrastructure, maintenance, and personnel costs), a service price, and a labor price for actual energy consumption. The price of district heating, if not based on fossil fuels, can therefore be assumed to be independent of gas or fossil fuel prices. I identify renewable district heating by its main energy source, excluding fossil fuel–based district heating networks. Of the 1,001 networks in the data, 256 are classified as renewable based on a fossil fuel share of at most 30%. The main energy sources for these renewable networks are waste incineration, geothermal energy, bioethanol, and biogas..

Treatment is defined by a large and unexpected increase in relative heating costs between gas-heated and renewable DH apartments, resulting from changes in gas prices after the Russian invasion in February 2022. Prior to 2022, consumer prices for gas and district heating move in parallel, as shown in Figure 1. Gas prices increase only slightly from 2020 to 2021 due to regulatory changes, including an increase in the VAT rate, higher network charges and the introduction of the Fuel Emissions Trading Act, i.e. national CO_2 certificate trading.[23] Toward the end of 2021, wholesale gas prices rise due to growing energy demand driven by the post-pandemic economic recovery, historically low gas storage levels in Germany, and a lack of supply increases from major gas exporters. For example, Gazprom refused to provide additional gas capacity and unplanned maintenance at a major Norwegian gas field led to lower

[22] By definition the main supplier has the largest market share and also serves as a default supplier.

[23] Refer to the German Federal Statistical Office (2021) for detailed information on these regulatory changes, including the Fuel Emission Allowance Trading Act (BEHG) (2020).

gas flows from Norway. However, average gas consumer prices remained constant until January 2022, when gas suppliers updated their prices to pass on higher import costs. The gas price increase in January 2022 is therefore a delayed pass-through of the higher wholesale price. The slight increase in average district heating prices is due to the share of gas-dependent district heating, which is accounted for in the following analysis by focusing on renewable district heating.

The Russian invasion in February 2022 is the key event that disrupted energy markets. On the one hand, political sanctions and reduced gas imports from Russia exacerbated the friction between gas supply and demand and put pressure on gas prices. Thus, the invasion dramatically intensified developments on the gas market, which could only be partially compensated by imports from Norway and the use of LNG, mainly from the US. On the other hand, the media attention on the limited gas supply and the low level of gas storage led to fears of insecure gas supply and gas rationing. This increased the salience of gas market developments, but may also have increased risk aversion and uncertainty about expected gas supply shortages. Figure A6 illustrates the unusually high attention to gas prices based on Google trend data for Germany. Moreover, the Boeck & Zörner (2025) finds that the gas price shock also impacted peoples' inflation expectations. Based on data from the German Heating and Housing Panel (Frondel et al. 2023), tenants expected a large increase in heating costs relative to 2021 not only in 2022 but also for 2023. Together this suggests that the gas price shock not only affected short-term heating cost expectations but also medium to long-term expectations. At last, several papers shows a direct effect of the energy prices shock in 2022 on overall inflation expectations (Wehrhöfer 2023, Patzelt & Reis 2024, Jo & Klopack 2025) suggesting a lasting effect on peoples' price expectations.

The observed decline in energy prices towards the end of 2022 was driven by large-scale policy interventions aimed at reducing residential heating costs. To avoid potential bias from these interventions, this study restricts the analysis period to the month before December 2022, before the announcement and implementation of the price control measures.

3.2.2 Specifications

I estimate the following TWFE regression and account for heterogeneous treatment effects across time:

$$\log(y_{jtr}) = \eta_t + \lambda_r + \sum_{k=-24, k\neq -1}^{9} \beta_k \mathbb{1}_{gas_j} \mathbb{1}_{T(t)=k} + \gamma_1 X_j + \gamma_2 X_{rt} + \epsilon_{jtr} \tag{5}$$

where y_{jtr} represents the rental price per sqm, time on market, or clicks per day of apartment j listed in month t located in zip code r. The dummy variable $\mathbb{1}_{gas_j}$ is one for all gas-heated apartments and zero for DH apartments. The dummy variable $\mathbb{1}_{T(t)=k}$ indicates month $T(t)$, k months after the treatment onset in February 2022. The parameters of interest β_k capture the difference in y_{jtr} between gas and DH apartments in month $T(t)$ relative to the reference month January 2022 ($k = -1$). For $k < 0$ the estimated pre-treatment differences allow testing for anticipation effects and for validation of the parallel trends assumption. For $k \geq 0$ the parameters capture period-specific treatment effects caused by an increase in heating costs, improved price salience, and supply uncertainty. λ_r and η_t capture year times month and zip code fixed effect, respectively.[24] X_j and X_{rt} include apartment and time-varying neighborhood-specific control variables.[25] Finally, ϵ_{jtr} is an identically and independently distributed

[24] Singleton observations are omitted in the high-dimensional fixed effect regressions to avoid incorrect inference and an overestimation of significance levels in the presence of fixed effects nested in clusters. For a more detailed discussion on the effect of singleton observations in high-dimensional fixed effects regressions refer to Correia (2015).

[25] Apartment specific control variables include (squ.) floor space, flat type FE, year of construction FE, condition FE, energy efficiency level, and type of landlord FE. Neighborhood controls at the one square kilometer grid level are purchasing power, population, number of houses, unemployment rate, and share of inhabitants aged 0-10, 11-18, 19-50, and 50 and older. When estimating the effect on time on market or clicks per day, the listed rental price per sqm is also included as covariate.

error term. The standard errors are clustered at the zip code level.

3.2.3 Identification

The fundamental identification assumption is that in the absence of treatment the difference in potential outcomes between the treatment and control group is constant over time. In the underlying setting, this implies that without an increase in the consumer price of gas, i.e., gas-based heating costs, the rental price, time on market, and clicks per day would have evolved identically for gas-heated and DH apartments. Figure A5 shows the unconditional trends in the outcomes of the treatment and control groups. Visual inspection suggests parallel trends before 2022. In addition, I test for significant pre-treatment differences based on equation (5). The estimates in Figure 5 support the evidence of parallel trends.

In the following, I discuss the three main threats to identification. First, the main concern is the credibility of the control group. Tenants must be able and willing to choose a DH apartment over a gas-heated apartment due to an increase in gas-based heating costs. To this end, gas-heated and DH apartments should be close substitutes with similar characteristics and spatial availability. I ensure substitutability by excluding neighborhoods without available renewable district heating or gas heating. This accounts for differences in the spatial availability of the two heating types and makes district heating a viable alternative. Table A1 shows the descriptive statistics for the used sample of gas-heated and renewable DH apartments before and after February 2022. I address remaining differences in apartment-specific characteristics in a robustness check using entropy balancing following Hainmueller & Xu (2013).[26]

Second, a potential risk to identification is measurement error in self-reported heating types and intentional misreporting of energy sources. In particular, landlords may not report an apartment as having gas heating during periods of higher gas prices in anticipation of lower demand for gas heating. Figure A1 shows the composition of all listed apartments by heating type over time. The composition remains constant until December 2022, suggesting no strategic misreporting of heating types during the observation period. Similarly, Figure 5(d) shows a common trend between the treatment and control groups with respect to the number of listed apartments. Both rejects misreporting of heating types or tenants moving out of gas-heated apartments due to higher heating costs.

Third, one might expect anticipation effects since agents might have already observed rising wholesale prices for gas in 2021 expecting an increase in consumer prices. Figure A6 presents some evidence of potential anticipation effects based increased attention to the gas prices on Google towards the end of 2021. Yet, Figure 5 displays no significant differences between treatment and control groups before January 2022. Moreover, it appears unlikely that tenants anticipated the Russian invasion in February 2022 and the associated disruptions on global energy markets.

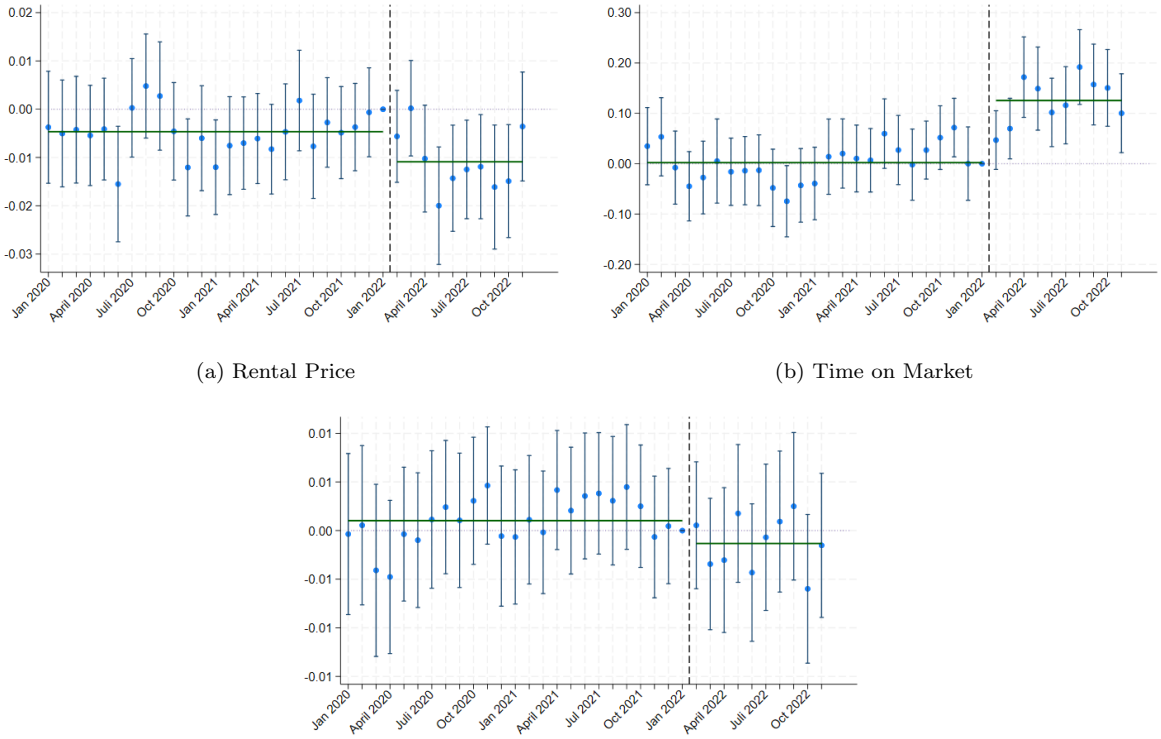
3.3 Results

Figure 5 presents the estimation results of equation (5), showing a small negative effect of heating costs on rental prices, a notably larger positive effect on time on market, and no statistically significant effect on listing clicks.

Compared to renewable DH apartments, rental prices of gas-heated apartments tend to be lower after January 2022. Estimated effects before April are very close to zero and only increase to 1-2% between April and October. This slightly delayed effect could be explained by increased treatment intensity with rising gas prices and the gradual implementation of sanctions over time, as well as a learning effect of agents that the conflict would last longer than just a few weeks, which changed their expectations. For

[26] For the continuous DiD design, I use the extension by Tübbicke (2022).

Figure 5: Impact of Heating Costs



(a) Rental Price

(b) Time on Market

(c) Clicks per day

Notes: The Figures plot the estimates and 95% confidence intervals from equation (5). Standard errors are clustered at the zip code level. The outcome variables are the log rental price per sqm, the log time on market in days, and the log clicks per day. The number of observations is 636,517 in all regressions and the R^2 is 0.831 in 5(a), 0.168 in 5(b), and 0.138 in 5(c).

example, the German government raised the alert level of its contingency plan at the end of March - around the same time that rental price differentials increase. Nevertheless, the estimated effects are relatively small and only borderline statistically different from the average pre-treatment effect indicated by the horizontal green line. The drop in rental price differentials in November may reflect anticipation of gas subsidies announced and implemented in December 2022, or easing concerns about limited gas storage and secure gas supplies for the upcoming heating season.

In contrast, the estimated effect on time on market is statistically significant and shows an average increase of 12.5% for gas-heated apartments after January 2022. At the peak, gas-heated apartments posted in August 2022 are listed 19.8% longer than renewable DH apartments, suggesting a significant decrease in housing demand due to increased heating costs. Relative to the average time on market, a decrease of 19.8% is equivalent to an additional 4-5 days. However, the used definition of time on market, i.e., listing duration, does not perfectly capture search duration. In fact, a more realistic interpretation of listing duration may be the time it takes a landlord to be contacted by a sufficient number of interested tenants. The actual time on market or search duration will therefore be longer, and the estimated effects on search duration will be underestimated in absolute terms. Again, the estimates show a delayed treatment effect after March and a decline to pre-treatment levels in November. The time dynamics of the estimated effect is thus similar to the effect on rental prices, but more pronounced and statistically significant. The estimated effect on clicks generated per day is statistically insignificant, but mostly negative and below pre-treatment levels.

For none of the three outcome variables, the estimation results show significant pre-treatment differ-

ences between gas-heated and renewable DH apartments. This supports the validity of the parallel trend assumption and the causality claim of the estimated parameters.

In the context of the model discussed in section 2, the estimates reject a capitalization of heating costs in rental prices. Increased heating cost only have a small effect on rental prices, suggesting a rather energy price-inelastic rental prices. Meanwhile, the increase in time on market for relatively more heating cost-intensive gas-heated apartments indicates an energy price sensitive demand for housing. Insignificant effects on clicks suggest that tenants do not change their submarket choice and do not filter for other key characteristics with increased gas prices.

3.4 Robustness

In this section, I conduct a series of robustness checks to test the validity of the identification assumptions discussed in section 3.2.3.

The main threat to identification is the adequacy of the control group, i.e., whether DH apartments are indeed close substitutes for gas-heated apartments with similar characteristics and local availability. I test the assumption of substitutability between the two apartment types in two ways. First, I apply an entropy balancing approach of Hainmueller & Xu (2013) to control for differences in apartment-specific characteristics. As shown in Table A1, the treatment and control group do indeed differ on average in some characteristics. For example, gas-heated apartments are generally older, less likely to be listed for first occupancy or by private landlords, and more energy efficient. The entropy weights are chosen to capture differences in the first, second, and third moments of the distribution of the control variables between the treatment and the control group. The weighted sample displays much smaller differences between gas-heated and DH apartments as shown in Table A2. The remaining average differences between treatment and control group are small and confirm a better comparability of both groups. Yet, the estimates are robust to entropy weighting (Figure A10) suggesting that the the baseline estimates are unbiased with respect to differences in apartment characteristics. Based on the weighted sample, the estimates confirm a positive effect of heating costs on time on market. As before, the estimated effect on rental prices is negative but statistically insignificant, and there is no effect on clicks.

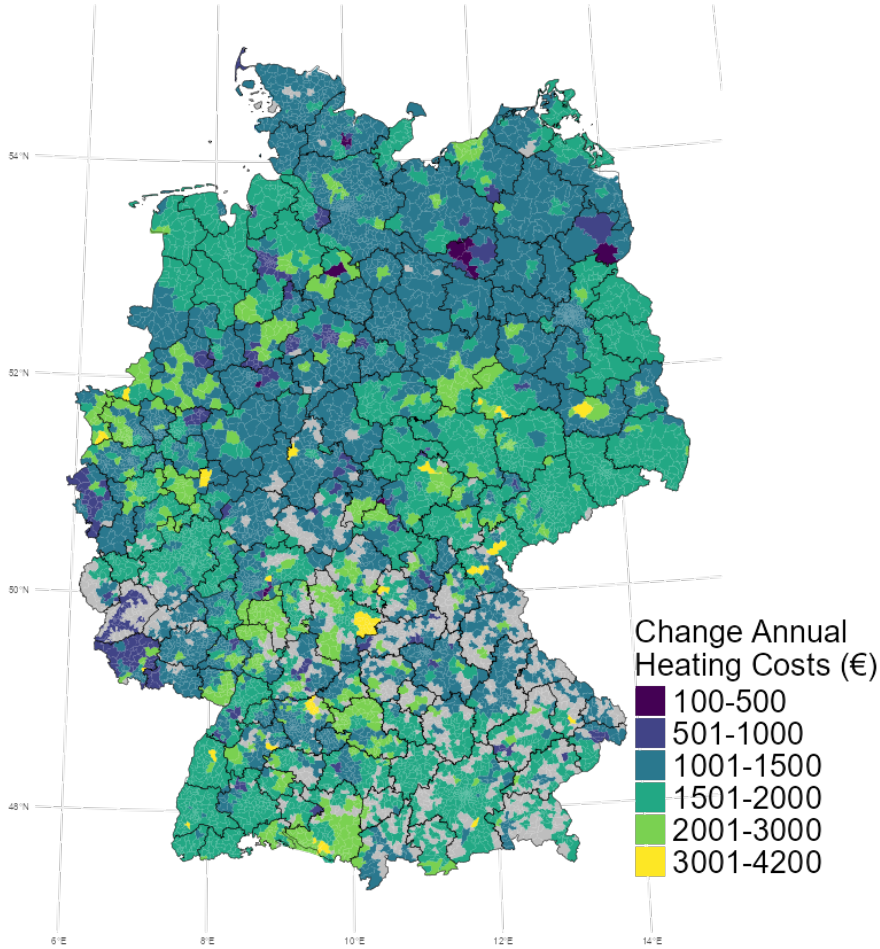
Second, I control for potentially different local availability of both apartment types at the zip code level. So far, the analysis is limited to zip code areas with both gas and renewable DH available, regardless of their respective market shares. To ensure that DH apartments are indeed a low-heating cost alternative to gas-heated apartments, I restrict the sample to zip code areas with at least 10% DH and gas-heated apartments, respectively. The estimated effects remain robust (Figure A11), suggesting that the results are not affected by a limited availability of DH apartments.

Finally, I address the potential threat of omitted variable bias and additionally control for (i) local labor market specific time trends to account for potentially heterogeneous housing market developments (Figure A12), (ii) one square kilometer grid cell fixed effects to more accurately capture time-invariant location specific effects (Figure A13), and (iii) a broader set of apartment specific control variables (Figure A14).[27] The results remain robust to all three specifications and confirm a positive effect on time on market and no significant effect on rental price and clicks.

In sum, the robustness checks confirm the baseline results and lend credence to the identification assumptions. Across all specifications, the positive effect on time on market persists, while effects on rent and clicks remain insignificant.

[27] Additional control variables include the presence of a balcony, elevator, garden, parking space, basement, number of bathrooms, and number of bedrooms if the apartment has a fitted kitchen.

Figure 6: Local Gas Cost Change



Notes: Displayed values capture the difference in average annual gas cost for a household with a yearly consumption of 20,000 kWh between 2020 and 03-11/2022 of the main supplier at the the zip code level. The black lines indicate the border of local labor markets following the definition of Kosfeld & Werner (2012).

3.5 Capitalization of Local Energy Prices

So far, the estimated effect of heating costs is motivated by differences in the average national consumer price for gas and district heating. However, the increase in gas prices affected neighborhoods to varying degrees. As shown in Figure 6, the change in annual gas costs for households supplied by the main supplier varied significantly across zip code areas. Even between adjacent zip code areas, annual gas costs changes can vary significantly and by several hundred euros. These price differences can be explained by main supplier-specific effects, as they were affected to different degrees depending on their procurement strategy, i.e., depending on the share of Russian gas and the duration of their import contracts. Especially for suppliers with a high reliance on Russian gas or very short-term procurement, an increase in global gas prices has a direct impact on consumer prices.

I estimate the impact of local gas price shocks based on equation (5) with a continuous treatment definition. Thus, the following TWFE regression estimates the period-specific effects of a marginal change in local gas-based heating costs (Δh_{rs}):

$$\log(y_{jtr}) = \eta_t + \lambda_r + \sum_{k=-24, k\neq -1}^{9} \beta_k \mathbb{1}_{gas_j} \mathbb{1}_{T(t)=k} \Delta h_{rs} + \gamma_1 X_j + \gamma_2 X_{rt} + \epsilon_{jtr} \tag{6}$$

where the change in local heating costs Δh_{rs} is defined by the absolute change in annual gas costs and the difference in average costs between January to August 2021 and March to November 2022 for a household with a consumption of 20,000 kwh. Figure A7 illustrates the distribution of local gas cost shocks, underlining large differences across zip codes. On average, annual gas costs increased by €1207.

The usage of a continuous treatment has crucial implication for identification and requires a stricter definition of the parallel trend assumption. In binary DiD designs, parallel trend relies on untreated potential outcomes of both treatment and control group. In continuous DiD designs, however, parallel trend relies on untreated potential outcomes across all levels of treatment intensities (Callaway et al. 2024). This so called *strong parallel trend assumption* rules out a correlation between treatment intensity and potential outcomes and therefore a selection into different levels of treatment intensity based on expected heating costs and restricts treatment effect heterogeneity.

To test the validity of the strong parallel trend assumption, I follow Havnes & Mogstad (2011) and Bauernschuster et al. (2016) and split the sample into two groups of zip code areas based on the magnitude of the local gas price increase. Figure A8 shows the unconditional evolution of the outcomes for the groups of below and above the median gas price increase. Based on this graphical assessment, zip code areas with a high and low gas price increase follow similar trends prior to 2022, suggesting that parallel trend also holds for different levels of treatment intensity.

Additionally, Figure A9 shows that the magnitude of the increase in local gas prices is uncorrelated with local pre-crisis gas prices. The gas price increase in 2022 did not reinforce already existing gas price differences and cannot be explained by local pre-treatment gas prices. This further supports the causality claim and absent selection into low treatment intensity.

Figure 7 shows the corresponding estimates and no significant effect of a marginal change in the local gas price on rental prices and clicks, but on time on market. An increase in annual heating costs by €1000 increases the time on market by 6%. At the peak, in April 2022, the estimated effect amounts to about 10%.

As in the case of the binary DiD design, the pre-treatment estimates for each of the three outcome variables are not statistically different from zero. Moreover, the results are robust to the series of robustness checks already discussed in section 3.4 (Figures A15-A19). Thus, the results confirm an absent heating cost capitalization in rental price and a heating cost elastic housing demand.

4 Discussion

The results presented in the previous section reject the capitalization of local heating costs in rental prices. Instead, the positive effect of heating costs on time on market suggests a decrease in housing demand due to increased relative heating costs and an energy price sensitive housing demand. In the context of the model discussed in section 2 this suggests a difference in the gas price elasticity of landlords and tenants.

Recall, the key feature of the model explaining the capitalization of heating costs in rental prices is that housing choices are based on total rental payments. A tenant is willing to rent an apartment only if the willingness to pay exceeds the sum of the rental price and the heating costs. Consequently, the landlord's optimization problem - as function of expected housing demand - also depends on both rental price and heating costs. The landlord considers that, given a fixed rental price, an increase in local heating costs reduces the transaction probability, since a smaller share of tenants is willing to rent. Only if housing demand is fully inelastic to changes in local heating costs, the equilibrium rental price is independent of heating costs. This requires that the probability of a transaction is independent of heating costs and that the tenant's decision to rent is fully determined by the rental price but not by

Figure 7: Impact of Local Energy Prices

(a) Rental Price

(b) Time on Market

(c) Clicks per day

Notes: The Figures plot the estimates and 95% confidence intervals from equation (6). Standard errors are clustered at the zip code level. The outcome variables are the log rental price per sqm, the log time on market in days, and the log clicks per day. The number of observations is 636,517 in all regressions and the R^2 is 0.831 in 7(a), 0.168 in 7(b), and 0.138 in 7(c).

the heating costs.

The results of this paper, however, show that an increase in gas-based heating costs does not affect rental prices, but prolongs the search of landlords of gas-heated apartments, i.e. a lower transaction probability at constant rental price. This significant effect of heating costs on time on market rejects a heating cost inelastic housing demand. Hence, the absence of heating cost capitalization in rental prices cannot be explained by an inelastic demand for housing. This is consistent with findings on price-elastic residential gas consumption, for example by Burns (2021), Favero & Grossi (2023), or Ahlvik et al. (2025).

A possible reason for the insignificant effect of heating costs on rental prices might be a low elasticity of housing demand. That is, landlords may not change the listed rental prices because the shift in housing demand, and thus the probability of a transaction, is too small. To rule this out, I conduct two heterogeneity analysis and test whether the effect of heating costs differs by local (i) housing market tightness or (ii) treatment intensity.

First, I show that the insignificant effect of heating costs on rental prices persists in less tight housing markets (i.e. less supply-dominated), where landlords should have greater incentives to adjust rental prices after a negative demand shock. In particular, I define tighter housing markets by an above-average population size since housing market are particular tight in large German cities. However, the estimated effect of heating costs does not differ by housing market tightness (Figure A20). In fact, the estimated effects are slightly larger for larger cities or tighter housing markets which contrast the notion of larger price responses in less tight housing markets.

Secondly, I test if the size of the shift in demand for gas-heated apartments impacts the effect on rental prices. To this end, I estimate the effect of heating costs for zip code areas with above and below

average gas cost increase. The corresponding estimates show no difference in the effect of heating costs between zip code areas with low and high treatment intensity (Figure A21). Yet, the results confirm a larger effect on time on market in zip code areas with an above average gas price increase. The size of the demand reduction appears not to affect the effect of heating costs on rental prices.

Taken together, I cannot find evidence that the lack of heating cost capitalization in rental prices is explained by a low demand elasticity or a too small reduction in demand. This supports the assumption that tenants' decisions to rent indeed depends on heating costs suggesting a negative relationship between heating costs and the transaction probability.

An alternative explanation is that landlords are less elastic to changes in heating costs as tenants - show differences in the perception parameter κ.[28] Since landlords do not bear the heating costs themselves, they may have less incentive to acquire relevant information (Sallee 2014, Angeletos & Sastry 2025), whereas tenants - who directly pay these costs - are more likely to be informed. This could explain why, on average, gas-heated apartments have tended to be listed at lower prices after February 2022 (Figures 5), while the marginal effect of changes in local gas costs appears to have no impact on rental prices (Figures 7). Although landlords may be aware of general trends in relative heating costs, the limited salience and higher cost of acquiring detailed local information can hinder capitalization. In this context, commercial landlords likely face fewer information frictions due to scale advantages and more professionalized operations, which make them more strategic and better informed — also with respect to local energy prices. This is supported by the evidence that only commercial landlords reduced rental prices for gas-heated apartments after February 2022, while private landlords did not (Figure A22). These findings reinforce the idea that information frictions play a key role in shaping the capitalization of heating costs into rental prices.

5 Conclusion

Given the high CO_2 emissions from residential heating, substantial efforts are required in the coming years to reduce the persistently high share of fossil-based heating and to achieve global climate goals. A solid understanding of the capitalization of energy prices in the housing market is therefore essential for the design of effective policies. Such policies must both encourage the transition to cleaner heating systems and improvements in energy efficiency, and address the impact of rising energy costs on households, e.g. through carbon pricing .

In this paper, I use the 2022 gas price increase as a natural experiment to examine the capitalization of heating costs in the housing market. Using a DiD framework, I compare gas-heated apartments to those using renewable district heating and estimate the impact of increased heating costs on rental prices and housing demand, measured by time on market and clicks. In addition, I exploit regional variation in gas price shocks to assess how local energy prices are capitalized into rental market outcomes.

The results show no evidence that rental prices reflect relative heating costs. However, higher heating costs have a significant impact on housing demand. Specifically, the 2022 gas price increase extends the time on market for gas-heated apartments by up to 19.8% relative to apartments with renewable DH, and by 12.5% on average. Similarly, marginal changes in local heating costs do not affect rental prices and clicks, but do affect time on market. An increase in local annual heating costs of €1,000 leads to an average increase in time on market of 6%, with effects reaching up to 10% in some months.

The results of this paper have important policy implications. On the one hand, housing demand appears to be sensitive to heating costs, supporting previous evidence that households are responsive to energy costs and that residential gas demand is energy price elastic (Alberini et al. 2020, Burns

[28] Rental prices are fully independent of heating costs for $\kappa^L = 0$.

2021, Favero & Grossi 2023, Rubin & Auffhammer 2024). On the other hand, despite this elasticity of demand, the results do not support the capitalization of relative heating costs in rental prices. This raises concerns about the effectiveness of carbon pricing and other cost-based incentives in promoting retrofitting or energy efficiency investments in the rental market. If landlords are not able to pass on investment costs - despite reduced heating costs - through higher rents, the economic incentive to make such investments in rental properties is reduced. However, it remains unclear whether this lack of capitalization results from inattentive pricing (Mensch 2022) or from information asymmetries (Myers 2020). If landlords cannot credibly communicate the benefits of energy efficiency investments to tenants, mandatory and reliable energy certification is needed to credibly signal energy-related cost savings and justify corresponding rent adjustments (e.g. Myers 2020, Myers et al. 2022).

At last, given that rental prices are energy price inelastic while housing demand is elastic, an expected increase in fossil energy prices can lead to sorting across heating types, where households that cannot avoid fossil fuel heating become more vulnerable to rising energy prices and price shocks. Understanding these distributional effects is crucial for designing effective policies that not only reduce emissions but also gain public acceptance.

References

Ahlvik, L., Kaariaho, T., Liski, M. & Vehviläinen, I. (2025), Household-level responses to the european energy crisis, Technical report, CEPR Discussion Paper. 3, 18

Alberini, A., Khymych, O. & Ščasnỳ, M. (2020), 'Responsiveness to energy price changes when salience is high: Residential natural gas demand in ukraine', *Energy Policy* **144**, 111534. 3, 19

Allcott, H. & Greenstone, M. (2012), 'Is there an energy efficiency gap?', *Journal of Economic perspectives* **26**(1), 3–28. 3

Amaral, F., Soufflet, M., Zdrzalek, J. & Zetzmann, S. (2024), Green signals: Energy efficiency and german housing markets, Technical report, Kiel Policy Brief. 2

Angeletos, G.-M. & Sastry, K. A. (2025), 'Inattentive economies', *Journal of Political Economy* **133**(7), 000–000. 8, 19

Aydin, E., Brounen, D. & Kok, N. (2020), 'The capitalization of energy efficiency: Evidence from the housing market', *Journal of Urban Economics* **117**, 103243. 2

Bauernschuster, S., Hener, T. & Rainer, H. (2016), 'Children of a (policy) revolution: The introduction of universal child care and its effect on fertility', *Journal of the European Economic Association* **14**(4), 975–1005. 17

Behr, S. M., Köveker, T. & Kücük, M. (2025), Understanding energy savings in a crisis: The role of prices and non-monetary factors, Technical report, DIW Discussion Papers. 3

Best, M. C. & Kleven, H. J. (2018), 'Housing market responses to transaction taxes: Evidence from notches and stimulus in the UK', *The Review of Economic Studies* **85**(1), 157–193. 3

Bishop, K. C. & Kiribrahim-Sarikaya, O. (2024), 'Energy-efficient investments in housing', *Regional Science and Urban Economics* **107**, 103994. 3

Boeck, M. & Zörner, T. O. (2025), 'Natural gas prices, inflation expectations, and the pass-through to euro area inflation', *Energy Economics* **141**, 108061. 12

Braakmann, N., Dursun, B. & Pickard, H. (2023), Energy price shocks and the demand for energy-efficient housing: Evidence from russia's invasion of ukraine, Technical report, IZA Discussion Papers. 3

Breidenbach, P. & Eilers, L. (2018), 'RWI-GEO-GRID: Socio-economic data on grid level', *Jahrbücher für Nationalökonomie und Statistik* **238**(6), 609–616. 10

Brounen, D. & Kok, N. (2011), 'On the economics of energy labels in the housing market', *Journal of Environmental Economics and Management* **62**(2), 166–179. 2

Burns, K. (2021), 'An investigation into changes in the elasticity of us residential natural gas consumption: A time-varying approach', *Energy Economics* **99**, 105253. 3, 18, 19

Callaway, B., Goodman-Bacon, A. & Sant'Anna, P. H. (2024), Difference-in-differences with a continuous treatment, Technical report, National Bureau of Economic Research. 17

Campagnolo, L. & De Cian, E. (2022), 'Distributional consequences of climate change impacts on residential energy demand across italian households', *Energy Economics* **110**, 106020. 3

Carroll, J., Denny, E., Lyons, R. C. & Petrov, I. (2024), 'Better energy cost information changes household property investment decisions: Evidence from a nationwide experiment', *Energy Economics* **139**, 107909. 3

Correia, S. (2015), 'Singletons, cluster-robust standard errors and fixed effects: A bad mix', *Technical Note, Duke University* **7**. 12

Davis, L. W. (2023), 'Evidence of a homeowner-renter gap for electric appliances', *The Energy Journal* **44**(4), 53–64. 3

Deng, Y., Li, Z. & Quigley, J. M. (2012), 'Economic returns to energy-efficient investments in the housing market: evidence from singapore', *Regional Science and Urban Economics* **42**(3), 506–515. 2

Diamond, P. A. (1982), 'Wage determination and efficiency in search equilibrium', *The Review of Economic Studies* **49**(2), 217–227. 4, 5

Dolls, M., Fuest, C., Krolage, C. & Neumeier, F. (2025), 'Who bears the burden of real estate transfer taxes? Evidence from the German housing market', *Journal of Urban Economics* **145**, 103717. 3

Edenhofer, O., Gerster, A., Gessner, J., Myers, E., Pahle, M. & Palmer, K. (2025), 'Targeted policies to break the deadlock on heating bans', *Nature Climate Change* pp. 1–3. 1

EIA (2020), 'Residential energy consumption survey (recs)', *U.S. Energy Information Administration, Office of Energy Demand and Integrated Statistics, Form EIA-457A of the 2020 Residential Energy Consumption Survey* . 1

Eichholtz, P., Kok, N. & Quigley, J. M. (2013), 'The Economics of Green Building', *The Review of Economics and Statistics* **95**(1), 50–63.
URL: *https://doi.org/10.1162/REST_a_00291* 2

Eurostat (2025*a*), 'Distribution of population by tenure status, type of household and income group'. 3

Eurostat (2025*b*), 'Share of renewable energy in gross final energy consumption by sector'. 1

Favero, F. & Grossi, L. (2023), 'Analysis of individual natural gas consumption and price elasticity: Evidence from billing data in italy', *Energy Economics* **118**, 106484. 3, 18, 20

Federal Cartel Office (2016), 'Marktmacht von Plattformen und Netzwerken', *B6-113/15* .
URL: *https://www.bundeskartellamt.de/SharedDocs/Publikation/DE/Berichte/Think-Tank-Bericht.html* 10

Fritzsche, C. & Vandrei, L. (2019), 'The german real estate transfer tax: Evidence for single-family home transactions', *Regional Science and Urban Economics* **74**, 131–143. 3

Frondel, M., Gerster, A., Kaestner, K., Krieg, M., Pahle, M., Schwarz, A., Singhal, P. & Sommer, S. (2023), Fdz data description: The german heating and housing panel (ghhp)-wave 1, Technical report, RWI Datenbeschreibung. 12

Fuel Emission Allowance Trading Act (BEHG) (2020).
URL: *https://www.bmuv.de/fileadmin/Daten$_B$MU/Download$_P$DF/Gesetze/behg$_e$n$_b$f.pdf* 11

Fuerst, F., McAllister, P., Nanda, A. & Wyatt, P. (2015), 'Does energy efficiency matter to home-buyers? An investigation of EPC ratings and transaction prices in England', *Energy Economics* **48**, 145–156.
URL: *https://www.sciencedirect.com/science/article/pii/S0140988314003296* 2

Fuerst, F. & Warren-Myers, G. (2018), 'Does voluntary disclosure create a green lemon problem? energy-efficiency ratings and house prices', *Energy Economics* **74**, 1–12. 2

Galvin, R. (2023), 'Do housing rental and sales markets incentivise energy-efficient retrofitting of western Germany's post-war apartments? Challenges for property owners, tenants, and policymakers', *Energy Efficiency* **16**(4), 25. 3

Genesove, D. & Han, L. (2012), 'Search and matching in the housing market', *Journal of Urban Economics* **72**(1), 31–45. 3, 4

Gerarden, T. D., Newell, R. G. & Stavins, R. N. (2017), 'Assessing the energy-efficiency gap', *Journal of Economic Literature* **55**(4), 1486–1525. 3

German Federal Statistical Office (2021), 'Household electricity price in the first half year of 2021 was 32.62 cent per kilowatt-hour', *Press release No. 466* .
URL: *https://www.destatis.de/EN/Press/2021/10/PE21$_4$66$_6$1243.html* 11

Gillingham, K., Harding, M. & Rapson, D. (2012), 'Split incentives in residential energy consumption', *The Energy Journal* **33**(2), 37–62. 1, 3

Gillingham, K. T. & Watten, A. (2024), 'How is rooftop solar capitalized in home prices?', *Regional Science and Urban Economics* **107**, 104006. 2

Guan, Y., Yan, J., Shan, Y., Zhou, Y., Hang, Y., Li, R., Liu, Y., Liu, B., Nie, Q., Bruckner, B. et al. (2023), 'Burden of the global energy price crisis on households', *Nature Energy* **8**(3), 304–316. 3

Hainmueller, J. & Xu, Y. (2013), 'Ebalance: A stata package for entropy balancing', *Journal of Statistical Software* **54**(7). 13, 15

Harjunen, O. & Liski, M. (2014), 'Not so myopic consumers-evidence on capitalization of energy technologies in a housing market'. 2, 3

Haurin, D. (1988), 'The duration of marketing time of residential housing', *Real Estate Economics* **16**(4), 396–410. 3

Havnes, T. & Mogstad, M. (2011), 'Money for nothing? Universal child care and maternal employment', *Journal of Public Economics* **95**(11-12), 1455–1465. 17

International Energy Agency (2021), *Renewables 2021.*
URL: *https://www.oecd-ilibrary.org/content/publication/6dcd2e15-en* 1

Jamissen, D., Vatne, J., Holz, F. & Neumann, A. (2024), 'The price elasticity of natural gas demand of small consumers in germany during the energy crisis 2022', *Energy Efficiency* **17**(8), 98. 3

Jo, Y. J. & Klopack, B. (2025), 'Fueling expectations: The causal impact of gas prices', *Journal of Monetary Economics* p. 103797. 12

Kahn, M. E. & Kok, N. (2014), 'The capitalization of green labels in the california housing market', *Regional Science and Urban Economics* **47**, 25–34. 2

Kholodilin, K. (2025), The impact of governmental regulations on housing market: Findings of a meta-study of empirical literature, Technical report, DIW Discussion Papers. 3

Kholodilin, K. A., Mense, A. & Michelsen, C. (2017), 'The market value of energy efficiency in buildings and the mode of tenure', *Urban Studies* **54**(14), 3218–3238. 2

Kopczuk, W. & Munroe, D. (2015), 'Mansion tax: The effect of transfer taxes on the residential real estate market', *American economic Journal: economic policy* **7**(2), 214–257. 3

Kosfeld, R. & Werner, A. (2012), 'German labour Markets - New delineation after the reforms of German district boundaries 2007-2011', *Raumforschung und Raumordnung* **70**, 49–64. 16

Krainer, J. (2001), 'A theory of liquidity in residential real estate markets', *Journal of Urban Economics* **49**(1), 32–53. 4

Krikser, T., Profeta, A., Grimm, S. & Huther, H. (2020), 'Willingness-to-pay for district heating from renewables of private households in germany', *Sustainability* **12**(10), 4129. 3

Kryzanowski, L., Wu, T. & Wu, Y. (2024), 'Impact of carbon prices on household behaviors in housing transactions', *Available at SSRN 4383805* . 3

Lambin, X., Schleich, J. & Faure, C. (2023), 'The energy efficiency gap in the rental housing market: It takes both sides to build a bridge', *The Energy Journal* **44**(1), 75–92. 3

Lang, G. & Lanz, B. (2021), 'Energy efficiency, information, and the acceptability of rent increases: A survey experiment with tenants', *Energy Economics* **95**, 105007. 1, 3

Löffler, M. & Siegloch, S. (2021), 'Welfare effects of property taxation', *ZEW-Centre for European Economic Research Discussion Paper* (21-026). 3

Lyu, X. (2024), 'Revisiting property tax capitalization', *Regional Science and Urban Economics* **108**, 104039. 3

Mensch, J. (2022), 'Screening inattentive buyers', *American Economic Review* **112**(6), 1949–1984. 20

Mense, A. (2018), 'The value of energy efficiency and the role of expected heating costs', *Environmental and Resource Economics* **71**(3), 671–701. 2

Millar, M. I. & White, R. M. (2024), 'Do residential property assessed clean energy (pace) financing programs affect local house price growth?', *Journal of Environmental Economics and Management* **124**, 102936. 3

Mortensen, D. T. (1982), 'Property rights and efficiency in mating, racing, and related games', *The American Economic Review* **72**(5), 968–979. 4, 5

Mortensen, D. T. & Pissarides, C. A. (1994), 'Job creation and job destruction in the theory of unem-

ployment', *The Review of Economic Studies* **61**(3), 397–415. 4, 5

Myers, E. (2019), 'Are home buyers inattentive? Evidence from capitalization of energy costs', *American Economic Journal: Economic Policy* **11**(2), 165–188. 1, 2

Myers, E. (2020), 'Asymmetric information in residential rental markets: Implications for the energy efficiency gap', *Journal of Public Economics* **190**, 104251. 3, 4, 20

Myers, E., Puller, S. L. & West, J. (2022), 'Mandatory energy efficiency disclosure in housing markets', *American Economic Journal: Economic Policy* **14**(4), 453–487. 1, 2, 3, 20

März, S., Stelk, I. & Stelzer, F. (2022), 'Are tenants willing to pay for energy efficiency? Evidence from a small-scale spatial analysis in Germany', *Energy Policy* **161**, 112753.
URL: *https://www.sciencedirect.com/science/article/pii/S0301421521006194* 2, 3

Patzelt, P. & Reis, R. (2024), *Estimating the rise in expected inflation from higher energy prices*, London School of Economics and Political Science. 12

Piazzesi, M. & Schneider, M. (2009), 'Momentum traders in the housing market: Survey evidence and a search model', *American Economic Review* **99**(2), 406–411. 4

Piazzesi, M., Schneider, M. & Stroebel, J. (2020), 'Segmented housing search', *American Economic Review* **110**(3), 720–759. 4

Pissarides, C. A. (1985), 'Short-run equilibrium dynamics of unemployment, vacancies, and real wages', *The American Economic Review* **75**(4), 676–690. 4, 5

Ramos, A., Gago, A., Labandeira, X. & Linares, P. (2015), 'The role of information for energy efficiency in the residential sector', *Energy Economics* **52**, S17–S29. 3

Rubin, E. & Auffhammer, M. (2024), 'Quantifying heterogeneity in the price elasticity of residential natural gas', *Journal of the Association of Environmental and Resource Economists* (2), 319–357. 3, 20

Ruokamo, E. & Sahari, A. (2025), 'Household heterogeneity in valuation of heating energy costs', *Energy Efficiency* **18**(2), 13. 2

RWI – Leibniz Institute for Economic Research & ImmobilienScout24 (2024), 'RWI Real Estate Data - Apartments for Rent (Version 11)'. 1, 10

Sahari, A. (2019), 'Electricity prices and consumers' long-term technology choices: Evidence from heating investments', *European Economic Review* **114**, 19–53. 3

Sallee, J. M. (2014), 'Rational inattention and energy efficiency', *The Journal of Law and Economics* **57**(3), 781–820. 8, 19

Sejas-Portillo, R., Moro, M. & Stowasser, T. (2025), 'The simpler the better? threshold effects of energy labels on property prices and energy efficiency investments', *American Economic Journal: Applied Economics* **17**(1), 41–89. 2

Shen, X., Liu, P., Qiu, Y., Patwardhan, A. & Vaishnav, P. (2021), 'Estimation of change in house sales prices in the united states after heat pump adoption', *Nature Energy* **6**(1), 30–37. 2

Sieger, L. & Weber, C. (2023), 'Inefficient markets for energy efficiency? The efficiency premium puzzle in the German rental housing market', *Energy Policy* **183**, 113819.
URL: *https://www.sciencedirect.com/science/article/pii/S0301421523004044* 2, 3

Singhal, P., Sommer, S., Kaestner, K. & Pahle, M. (2025), 'Split-incentives in energy efficiency investments? Evidence from rental housing', *Resource and Energy Economics* p. 101488. 3

Slemrod, J., Weber, C. & Shan, H. (2017), 'The behavioral response to housing transfer taxes: Evidence from a notched change in dc policy', *Journal of Urban Economics* **100**, 137–153. 3

Stechemesser, A., Koch, N., Mark, E., Dilger, E., Klösel, P., Menicacci, L., Nachtigall, D., Pretis, F., Ritter, N., Schwarz, M. et al. (2024), 'Climate policies that achieved major emission reductions: Global evidence from two decades', *Science* **385**(6711), 884–892. 1

Taruttis, L. & Weber, C. (2022), 'Estimating the impact of energy efficiency on housing prices in germany:

Does regional disparity matter?', *Energy Economics* **105**, 105750. 2

Thiel, P. (2024), Fdz data description: Real-estate data for germany (rwi-geo-red v11)-advertisements on the internet platform lmmobilienscout24 2007-06/2024, Technical report, RWI Datenbeschreibung. 10

Tübbicke, S. (2022), 'Entropy balancing for continuous treatments', *Journal of Econometric Methods* **11**(1), 71–89. 13

Wehrhöfer, N. (2023), Energy prices and inflation expectations: Evidence from households and firms, Technical report, Deutsche Bundesbank. 12

Wheaton, W. C. (1990), 'Vacancy, search, and prices in a housing market matching model', *Journal of Political Economy* **98**(6), 1270–1292. 4

Williams, J. & Capital, P. (2014), Housing markets with construction, screening, and focused search, Technical report, Working paper. 5

Yoon, T., Ma, Y. & Rhodes, C. (2015), 'Individual heating systems vs. district heating systems: What will consumers pay for convenience?', *Energy Policy* **86**, 73–81. 3

Zheng, S., Wu, J., Kahn, M. E. & Deng, Y. (2012), 'The nascent market for "green" real estate in beijing', *European Economic Review* **56**(5), 974–984. 2

Zhu, Y., Zheng, Y. & Ren, Z. (2024), 'Household welfare loss from energy price crisis: Evidence from china', *Energy Economics* **138**, 107836. 3

A Appendix

A.1 Landlord Optimization Problem

In each period, a landlords is matched with a tenants with probability $q_m = q_m(\theta_m)$ which is a function of local housing market tightness in submarket m. In case of a match, the tenant does not accept the rental price with probability $G(p_j) = G_{V|F}\left(p_j + \kappa h_{rs}|f\right)$. Hence, a landlord receives the rental price p_j from period one onwards with probability $q_m\big(1 - G(p_j)\big)$. In case of no transaction in period one, with probability $1 - q_m\big(1 - G(p_j)\big)$ the game repeats. For an infinite time horizon the landlords expected lifetime utility can be derived as follows:

$$
\begin{aligned}
\max_p U(p_j) =& \overbrace{q_m\big(1 - G(p_j)\big)\sum_{t=0}^{\infty}\beta^t p_j}^{t=0} + \overbrace{q_m\big(1 - G(p_j)\big)\big(q_m G(p_j) + 1 - q_m\big)^1 \sum_{t=1}^{\infty}\beta^t p_j}^{t=1} + \dots \\
=& q_m\big(1 - G(p_j)\big)\sum_{k=0}^{\infty}\big(q_m G(p_j) + 1 - q_m\big)^k \sum_{t=k}^{\infty}\beta^t p_j \\
=& q_m\big(1 - G(p_j)\big)p_j \sum_{k=0}^{\infty}\big(q_m G(p_j) + 1 - q_m\big)^k \left(\frac{\beta^k}{1-\beta}\right) \\
=& p_j q_m \frac{1 - G(p_j)}{1-\beta}\sum_{k=0}^{T}\Big(\big(q_m G(p_j) + 1 - q_m\big)\beta\Big)^k \\
=& p_j q_m \frac{1 - G(p_j)}{1-\beta}\frac{1}{1 - (q_m G(p_j) + 1 - q)\beta} \\
=& p_j q_m \frac{1 - G(p_j)}{1 - \beta - (q_m G(p_j) + 1 - q)(\beta - \beta^2)}
\end{aligned}
$$

A.2 Landlord Optimization Problem with Limited Contract Duration

In this section, I present a modification of the standard search-and-matching model introduced in Section 2, incorporating limited rental durations through a Calvo-type pricing mechanism.

Previously, the standard model assumed that once a match between a landlord and a tenant is formed, the lease continues indefinitely. However, such an assumption is unrealistic. Instead, I assume that leases terminate each period with probability ω. Consequently, the probability that a lease lasts exactly t periods is $(1-\omega)^{t-1}\omega$.

The landlord's maximization problem now becomes:

$$\begin{aligned}
\max_p U(p_j) =& q_m\big(1-G(p_j)\big)\sum_{k=0}^{\infty}\big(q_m G(p_j)+1-q_m\big)^k\sum_{t=k}^{\infty}\beta^t\omega(1-\omega)^{t-1}p_j \\
=& p_j q_m\big(1-G(p_j)\big)\frac{\omega}{1-\omega}\sum_{k=0}^{\infty}\Big[\big(q_m G(p_j)+1-q_m\big)\big(\beta(1-\omega)\big)\Big]^k \\
=& p_j q_m\frac{\omega}{1-\omega}\frac{1-G(p_j)}{1-\beta(1-\omega)-(q_m G(p_j)+1-q)(\beta(1-\omega)-\beta^2(1-\omega)^2)}
\end{aligned}$$

The resulting optimal rental price under limited rental duration is:

$$\tilde{p}_j^* = (\bar{v}_m - \kappa h_{rs})\cdot\frac{1-\sqrt{1-\beta q_m(1-\omega)\big(1-\beta(1-\omega)(1-q_m)\big)}}{\beta(1-\omega)q_m} = \bar{x}_m\tilde{\Phi}_m \tag{7}$$

where $\tilde{\Phi}_m < \Phi_m$, implying that the optimal price is lower when landlords account for the possibility of tenants terminating the lease. Intuitively, this is because search costs become more significant when leases are shorter on average, as they are incurred more frequently. This results than landlords aiming to reduce time on market by lowering the asking price.

Tables

Table A1: Comparison of Treatment and Control Groups

	Pre-Treatment		**Post-Treatment**	
	Untreated	Treated	Untreated	Treated
Apartment Characteristics				
Base Rent per sqm (€)	11.05	10.29	11.83	10.98
Clicks per Day	0.08	0.06	0.08	0.07
Time on Market (Days)	24.71	22.89	21.56	21.69
Floor Space (m^2)	64.65	65.35	63.76	65.02
Number of Rooms	2.40	2.40	2.35	2.38
Energy Efficiency Score	94.36	123.09	94.38	121.07
Energy Efficiency Missing	0.23	0.38	0.25	0.39
Year of Construction	1980	1960	1980	1962
Year of Last Renovation	2011	2011	2010	2010
Commercial Landlord	0.81	0.69	0.78	0.67
First Occupancy	0.12	0.03	0.09	0.02
Condition: Like New	0.14	0.08	0.15	0.08
Condition: Renovated	0.15	0.21	0.15	0.22
Condition: Good	0.21	0.30	0.24	0.31
Commercial Provider	0.80	0.69	0.78	0.67
Flat Type: Attic	0.06	0.14	0.06	0.13
Flat Type: Basic	0.64	0.53	0.64	0.53
Flat Type: Hochparterre	0.02	0.03	0.02	0.03
Flat Type: Penthouse/Maisonette	0.02	0.04	0.02	0.04
Flat Type: Souterrain	0.01	0.01	0.01	0.01
Flat Type: With Terrace	0.03	0.03	0.04	0.04
Neighborhood Characteristics				
Population	6625	6142	6554	6107
Number of Buildings	669	756	718	797
Purchasing Power (mio €)	155.86	148.86	163.85	157.86
Unemployment Rate	8.13	7.59	8.16	7.61
Share Pop. Aged 10≥	9.38	9.44	9.45	9.51
Share Pop. Aged 11-18	6.54	6.72	6.66	6.81
Share Pop. Aged 52≤	39.71	39.87	39.46	39.90
Vacancy Rate of Apartments	4.37	4.39	4.43	4.28
Number of Observations	185,624	317,467	47,762	85,664

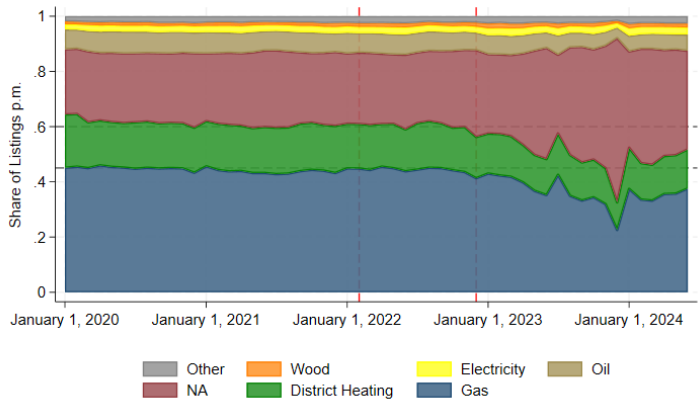
Note: All differences in means are statistically significant for both samples due to a large sample size. The means of the indicator variables for missing energy efficiency, first occupancy, private landlord, facility conditions and flat type represent the share of the sample. Neighborhood characteristics are at the one square kilometer grid cell level.

Table A2: Comparison of Treatment and Control Groups (Weighted vs. Unweighted)

	Unweighted		**Weighted**	
	Untreated	Treated	Untreated	Treated
Apartment Characteristics				
Base Rent per sqm (€)	11.21	10.45	10.78	10.45
Clicks per Day	0.08	0.06	0.07	0.06
Time on Market (Days)	24.07	22.64	22.14	22.64
Floor Space (m^2)	64.47	65.27	65.28	65.27
Number of Rooms	2.39	2.40	2.40	2.40
Energy Efficiency Score	94.36	122.67	119.22	122.67
Energy Efficiency Missing	0.24	0.38	0.38	0.38
Year of Construction	1980	1960	1965	1960
Last Renovation (Year)	2011	2011	2012	2011
Private Provider	0.80	0.69	0.69	0.69
First Occupancy	0.11	0.03	0.06	0.03
Condition: Like New	0.14	0.08	0.08	0.08
Condition: Renovated	0.15	0.21	0.21	0.21
Condition: Good	0.22	0.30	0.30	0.30
Flat Type: Attic	0.06	0.14	0.14	0.14
Flat Type: Basic	0.64	0.53	0.54	0.53
Flat Type: Hochparterre	0.02	0.03	0.03	0.03
Flat Type: Penthouse/Maisonette	0.02	0.04	0.04	0.04
Flat Type: Souterrain	0.01	0.01	0.01	0.01
Flat Type: With Terrace	0.03	0.04	0.04	0.04
Number of Observations	233,386	403,131	233,386	403,131

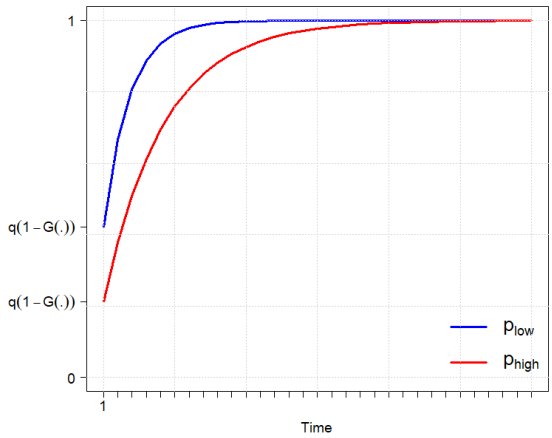
Figures

Figure A1: Composition of Heating Types

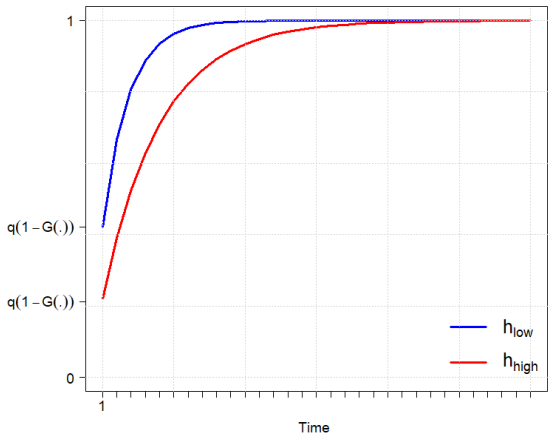


Notes: The graph displays the distribution of all listings in the period 01/2020 to 06/2024 by heating types. Recall, the observation period for the analysis is restricted to 11/2022. The onset of treatment is 02/2022.

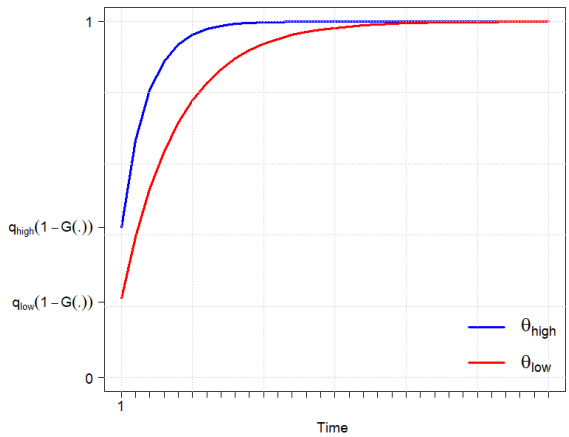
Figure A2: Cumulative Transaction Probability



(a) by Rental Price

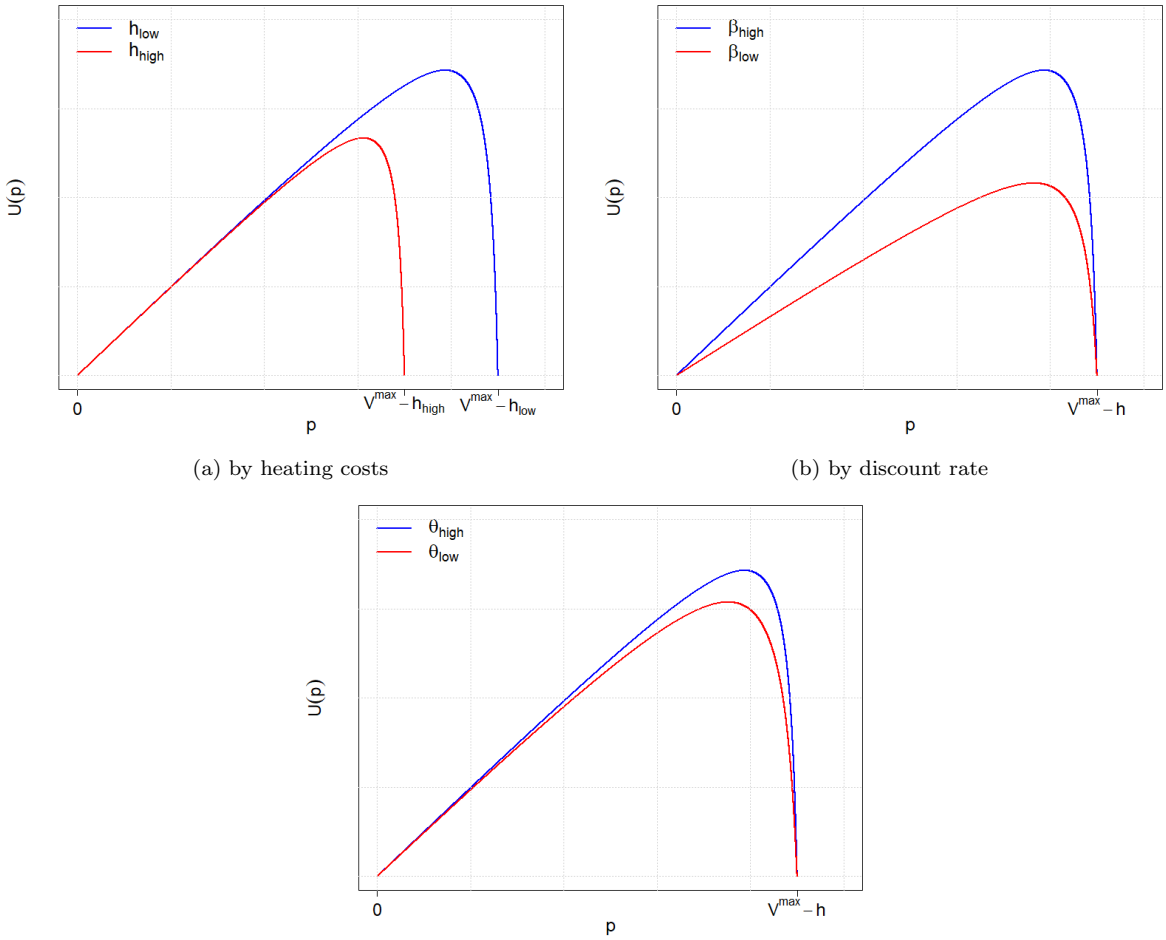


(b) by Heating costs



(c) Housing Market Tightness

Figure A3: Landlord Expected Lifetime Utility

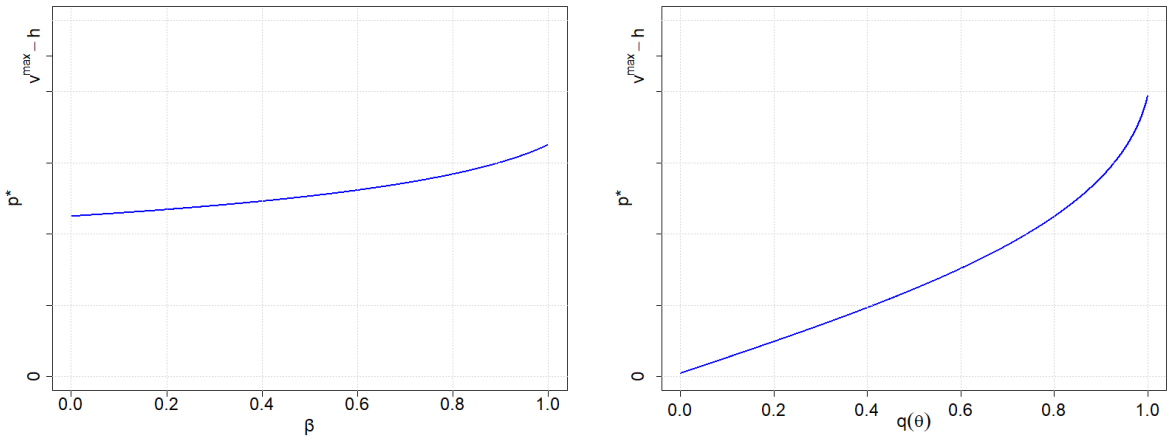


(a) by heating costs

(b) by discount rate

(c) by matching probability

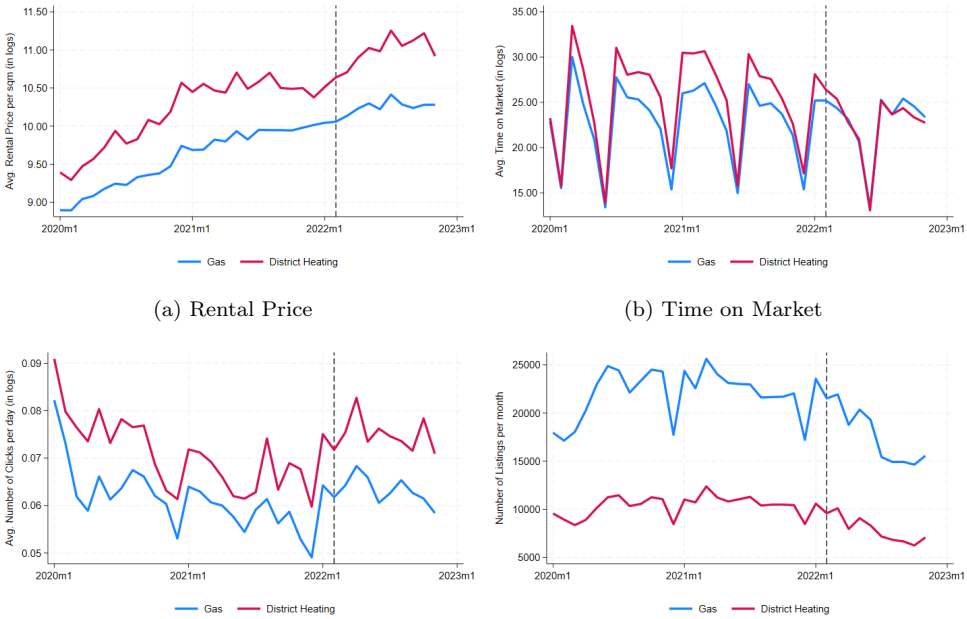
Figure A4: Utility Maximizing Rental Price



(a) by discount rate

(b) by matching probability

Figure A5: Common Trends



(a) Rental Price

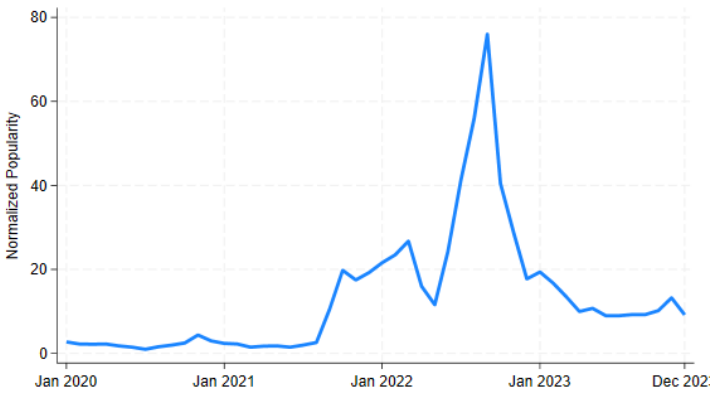
(b) Time on Market

(c) Clicks per day

(d) Listings

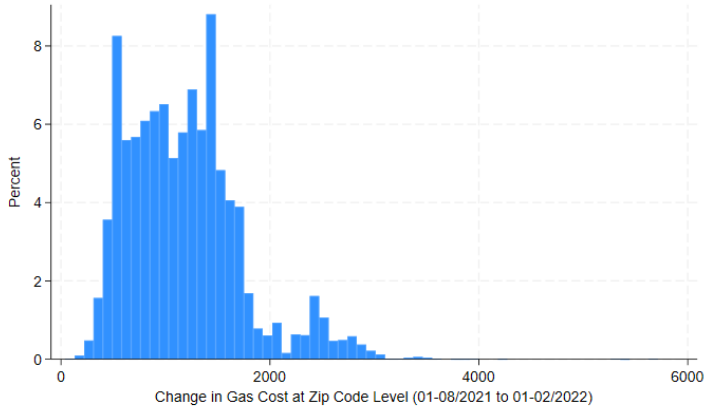
Notes: The graphs represent the common trend assumption of the DiD framework for the outcome variables rental price per sqm (5(a)), time on market (5(b)), and clicks per day (5(c)). Additionally, Figure 5(d) displays common trends for listings to illustrate the absence diverging posting trends.

Figure A6: Google Trend of "Gas Prices"



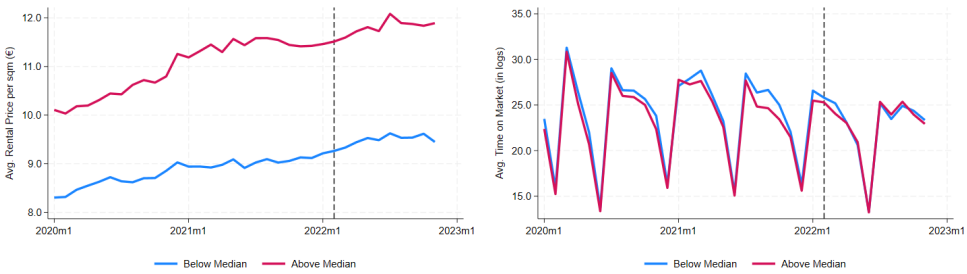
Notes: The graph shows the *Google Trends* on the number of times (normalized popularity) people in Germany have google *"Gaspreis"*, which translates to gas prices.

Figure A7: Distribution of local Gas Cost Changes



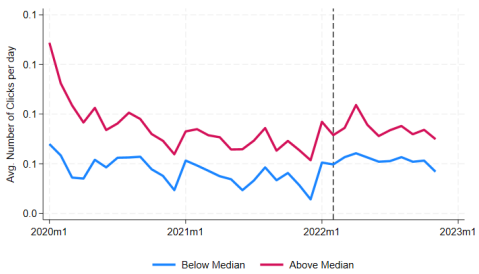
Notes: The histogram displays distribution in the change in annual gas cost across zip code areas for a representative household with a yearly consumption of 20,000 kWh.

Figure A8: Common Trends by Treatment Intensity



(a) Rental Price

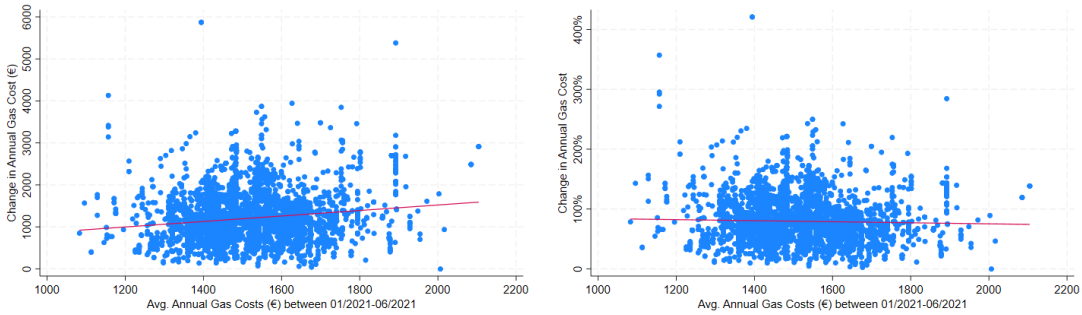
(b) Time on Market



(c) Clicks per day

Notes: The graphs represent the common trend assumption of the continuous DiD framework for the outcome variables rental price (8(a)), time on market (8(b)), and clicks (8(c)). The two groups describe zip code exposed to a below and above median increase in gas costs between 2020 and 2022.

Figure A9: Annual Gas Cost Increase

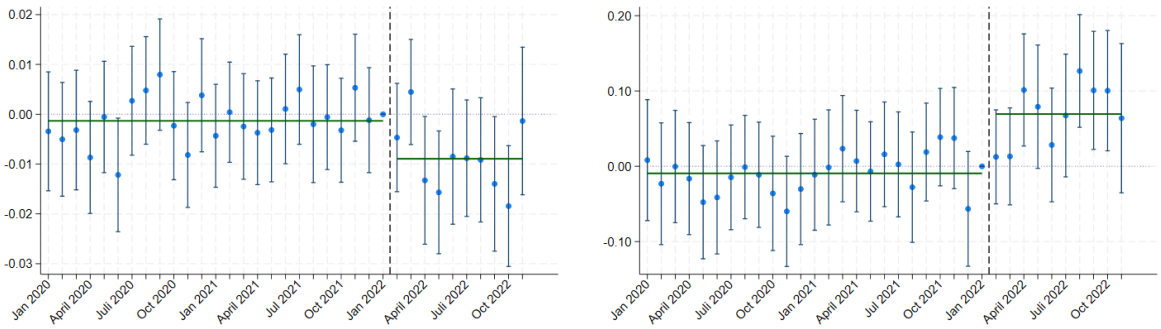


(a) Absolute Change (€)

(b) Percentage Change (%)

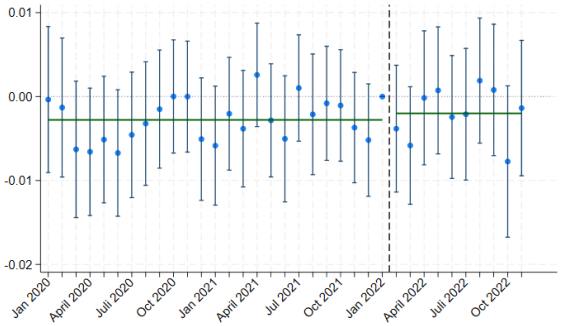
Notes: The graphs show the correlation of the change in annual gas costs and the difference in average costs between January to August 2021 and March to November 2022 with the level of gas costs in 2020 at the zip code level. Figure 9(a) shows the correlation for the absolute increase and Figure 9(b) for the relative increase.

Figure A10: Impact of Heating Costs - Balanced Sample



(a) Rental Price

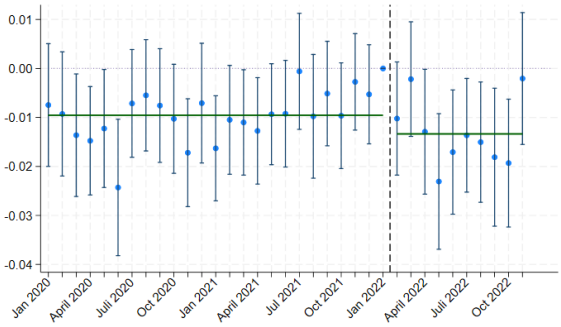
(b) Time on Market



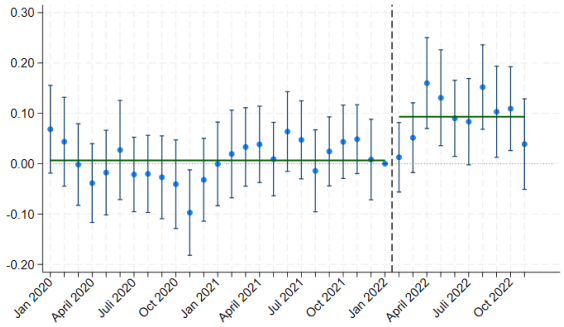
(c) Clicks per day

Notes: The figures plot the estimates and 95% confidence intervals from equation (5). Standard errors are clustered at the zip code level. The outcome variables are the log rental price per sqm, the log time on market in days, and the log clicks per day. The number of observations is 636,517 in all regressions and the sample is weighted using entropy balancing. The R^2 is 0.832 in 10(a), 0.168 in 10(b), and 0.138 in 10(c).

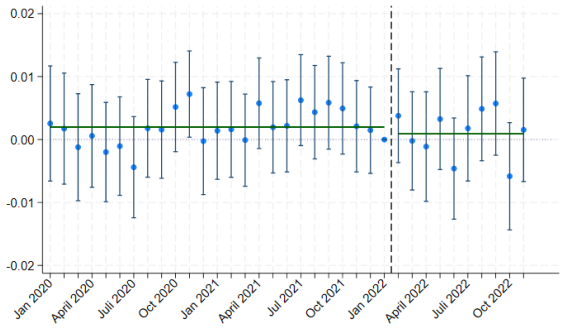
Figure A11: Impact of Heating Costs - Stricter Spatial Overlap



(a) Rental Price



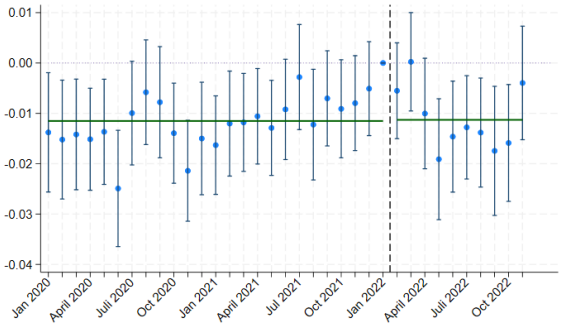
(b) Time on Market



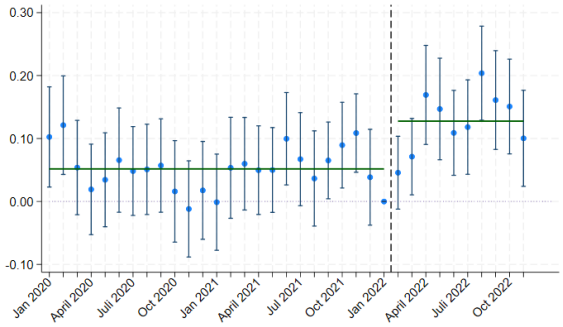
(c) Clicks per day

Notes: The figures plot the estimates and 95% confidence intervals from equation (5). Standard errors are clustered at the zip code level. The outcome variables are the log rental price per sqm, the log time on market in days, and the log clicks per day. The number of observations is 442,411 in all regressions and the sample is restricted to zip code areas with a share of at least 10% gas heated and DH apartments, respectively. The R^2 is 0.824 in 11(a), 0.168 in 11(b), and 0.143 in 11(c).

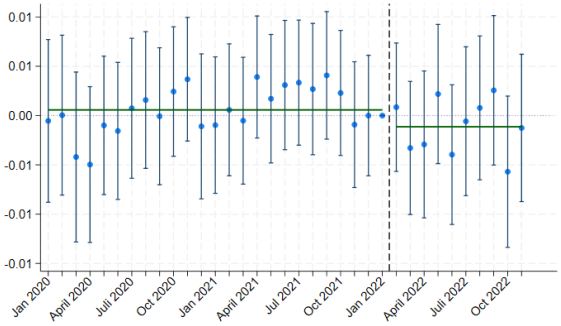
Figure A12: Impact of Heating Costs - LLM time trends



(a) Rental Price



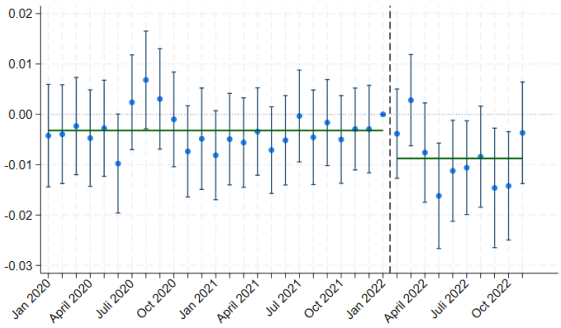
(b) Time on Market



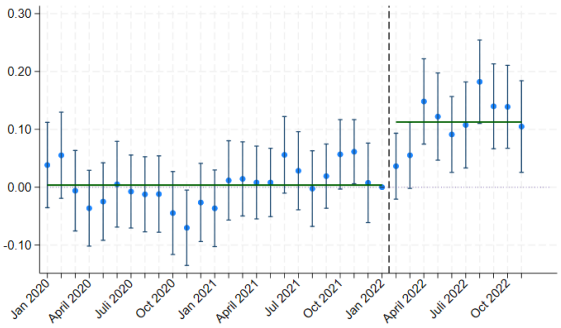
(c) Clicks per day

Notes: The figures plot the estimates and 95% confidence intervals from equation (5) also including local labor market specific time trends. Standard errors are clustered at the zip code level. The outcome variables are the log rental price per sqm, the log time on market in days, and the log clicks per day. The number of observations is 636,517 in all regressions and the R^2 is 0.832 in 12(a), 0.168 in 12(b), and 0.138 in 12(c).

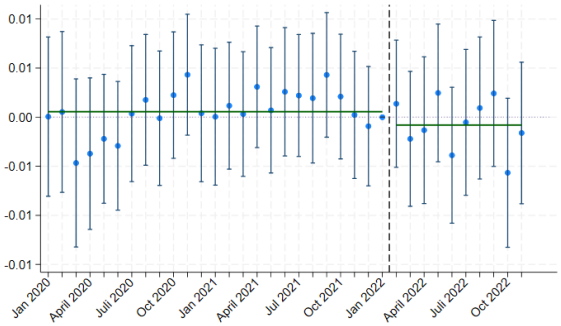
Figure A13: Impact of Heating Costs - 1 km^2 Grid Cell FE



(a) Rental Price



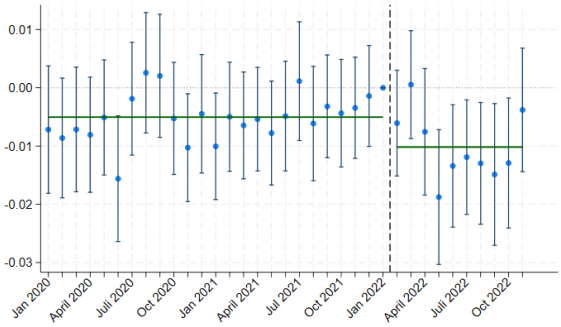
(b) Time on Market



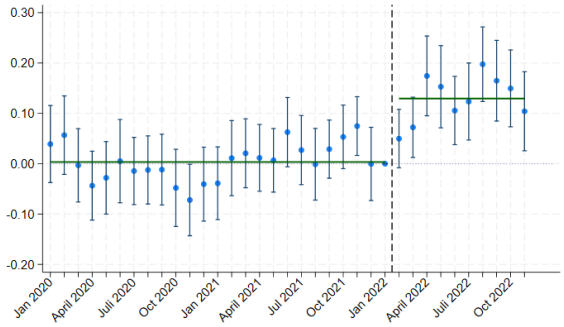
(c) Clicks per day

Notes: The figures plot the estimates and 95% confidence intervals from equation (5) with $1km^2$ grid cell fixed effects instead of zip code fixed. Standard errors are clustered at the zip code level. The outcome variables are the log rental price per sqm, the log time on market in days, and the log clicks per day. The number of observations is 636,517 in all regressions and the the R^2 is 0.832 in 13(a), 0.168 in 13(b), and 0.138 in 13(c).

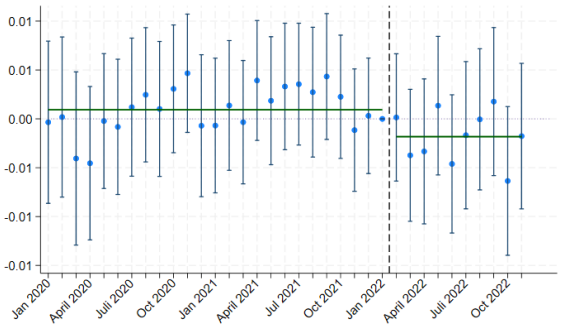
Figure A14: Impact of Heating Costs - Additional Control Variables



(a) Rental Price



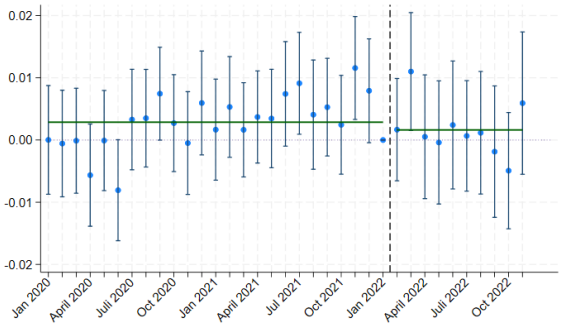
(b) Time on Market



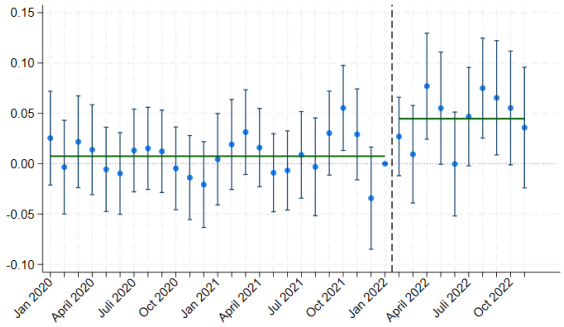
(c) Clicks per day

Notes: The figures plot the estimates and 95% confidence intervals from equation (5) with a more extensive list of control variables. Standard errors are clustered at the zip code level. The outcome variables are the log rental price per sqm, the log time on market in days, and the log clicks per day. The number of observations is 636,517 in all regressions and the R^2 is 0.832 in 14(a), 0.168 in 14(b), and 0.138 in 14(c).

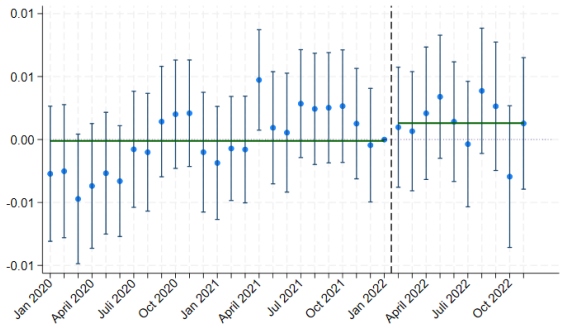
Figure A15: Impact of local Energy Prices - Balanced Sample



(a) Rental Price



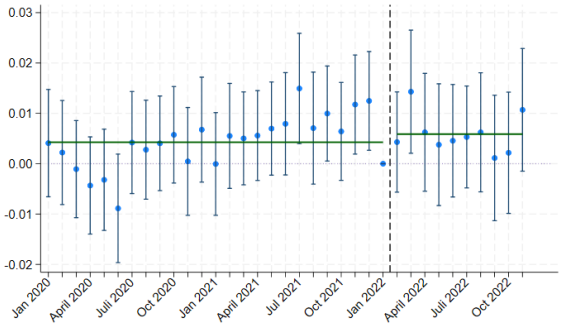
(b) Time on Market



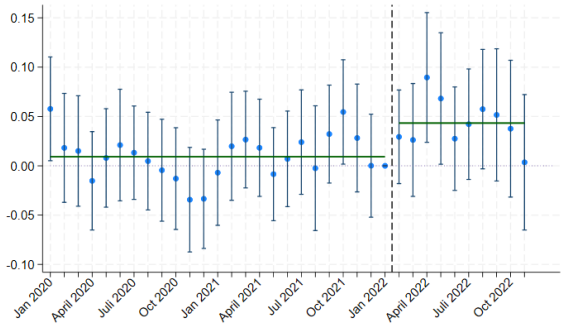
(c) Clicks per day

Notes: The figures plot the estimates and 95% confidence intervals from equation (5). Standard errors are clustered at the zip code level. The outcome variables are the log rental price per sqm, the log time on market in days, and the log clicks per day. The number of observations is 636,517 in all regressions and the sample is weighted using entropy balancing. The R^2 is 0.834 in 15(a), 0.168 in 15(b), and 0.143 in 15(c).

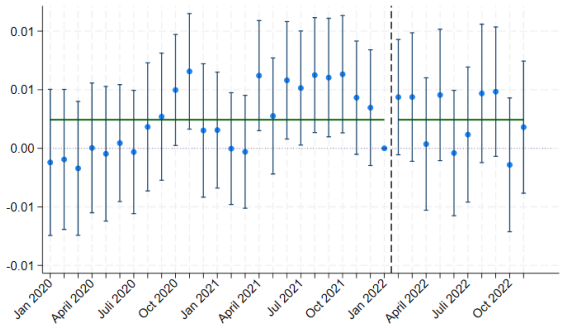
Figure A16: Impact of local Energy Prices - Stricter Spatial Overlap



(a) Rental Price



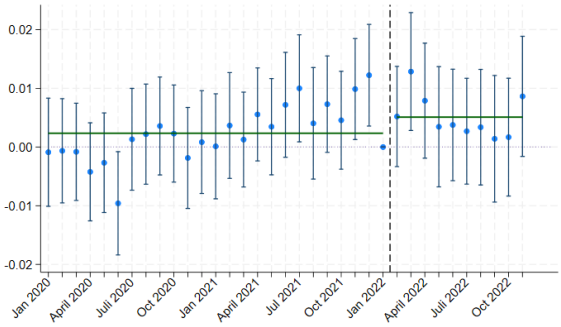
(b) Time on Market



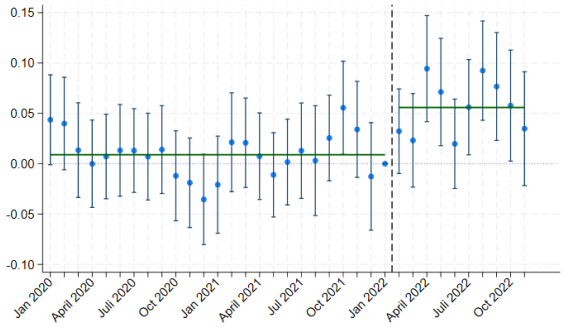
(c) Clicks per day

Notes: The figures plot the estimates and 95% confidence intervals from equation (5). Standard errors are clustered at the zip code level. The outcome variables are the log rental price per sqm, the log time on market in days, and the log clicks per day. The number of observations is 442,411 in all regressions and the sample is restricted to zip code areas with a share of at least 10% gas heated and DH apartments, respectively. The R^2 is 0.832 in 16(a), 0.168 in 16(b), and 0.138 in 16(c).

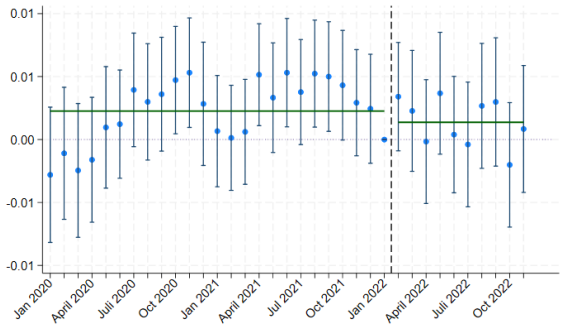
Figure A17: Impact of local Energy Prices - LLM time trends



(a) Rental Price



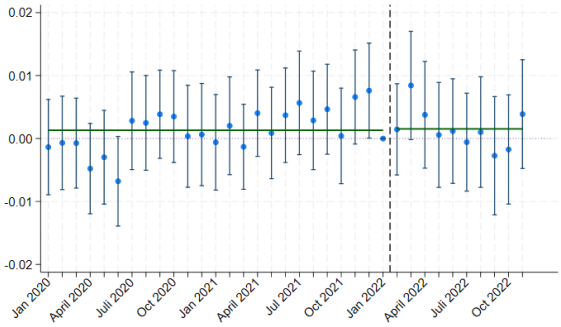
(b) Time on Market



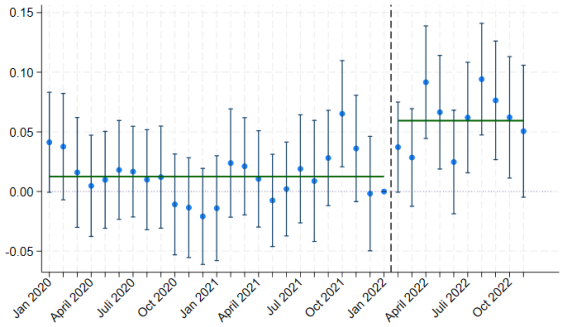
(c) Clicks per day

Notes: The figures plot the estimates and 95% confidence intervals from equation (5) also including local labor market specific time trends. Standard errors are clustered at the zip code level. The outcome variables are the log rental price per sqm, the log time on market in days, and the log clicks per day. The number of observations is 636,517 in all regressions and the R^2 is 0.832 in 17(a), 0.168 in 17(b), and 0.138 in 17(c).

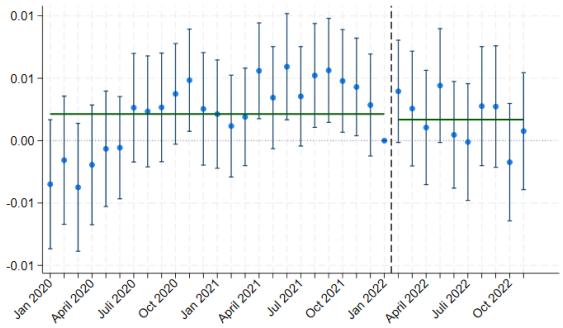
Figure A18: Impact of local Energy Prices - 1 km^2 Grid Cell FE



(a) Rental Price



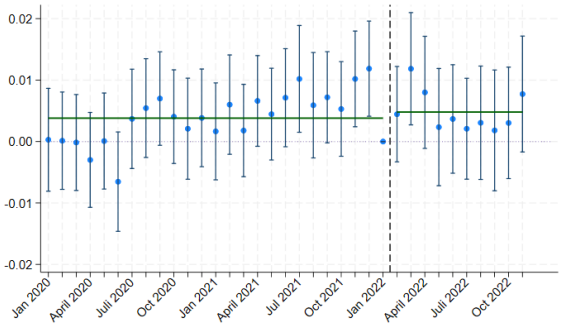
(b) Time on Market



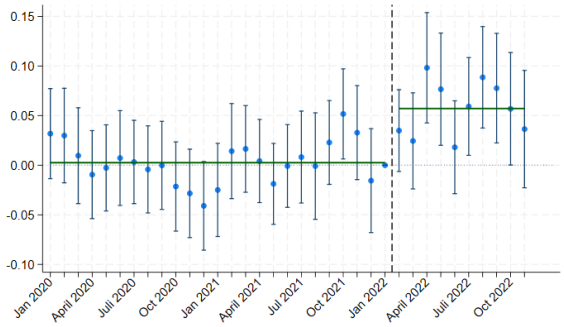
(c) Clicks per day

Notes: The figures plot the estimates and 95% confidence intervals from equation (5) with 1km2 grid cell fixed effects instead of zip code fixed. Standard errors are clustered at the zip code level. The outcome variables are the log rental price per sqm, the log time on market in days, and the log clicks per day. The number of observations is 636,517 in all regressions and the R^2 is 0.832 in 18(a), 0.168 in 18(b), and 0.138 in 18(c).

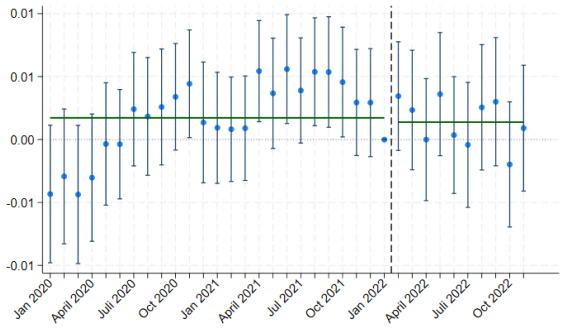
Figure A19: Impact of local Energy Prices - Additional Control Variables



(a) Rental Price



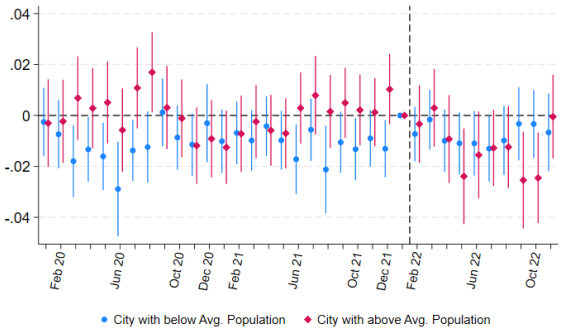
(b) Time on Market



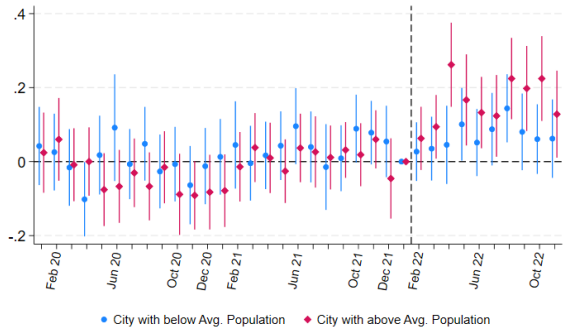
(c) Clicks per day

Notes: The figures plot the estimates and 95% confidence intervals from equation (5) with a more extensive list of control variables. Standard errors are clustered at the zip code level. The outcome variables are the log rental price per sqm, the log time on market in days, and the log clicks per day. The number of observations is 636,517 in all regressions and the R^2 is 0.832 in 19(a), 0.168 in 19(b), and 0.138 in 19(c).

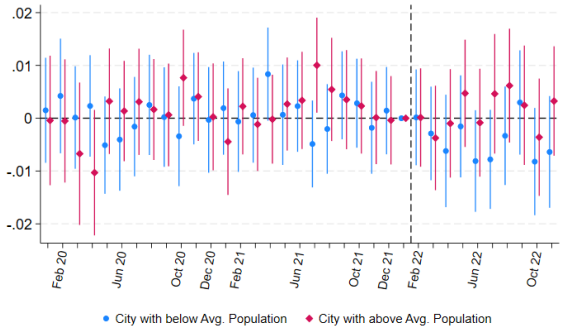
Figure A20: Impact of Heating Costs by City Size



(a) Rental Price



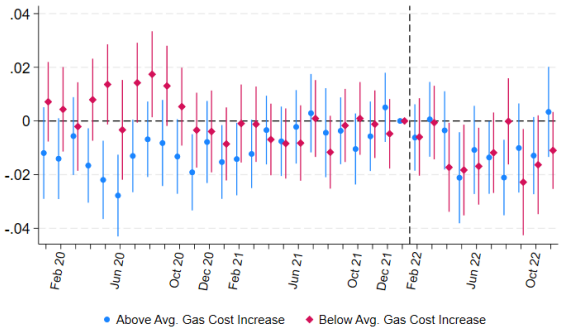
(b) Time on Market



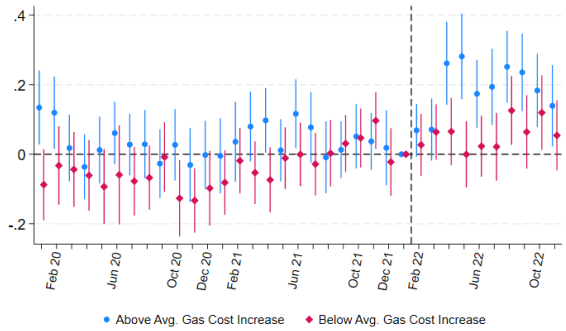
(c) Clicks per day

Notes: The Figures plot the estimates and 95% confidence intervals from equation (5) for below and above average population cities. Here, city size serves as a proxy for housing market tightness. Standard errors are clustered at the zip code level. The outcome variables are the log rental price per sqm, the log time on market in days, and the log clicks per day. The number of observations is 332,938 for the below-average sample and 303,579 for the above-average sample.

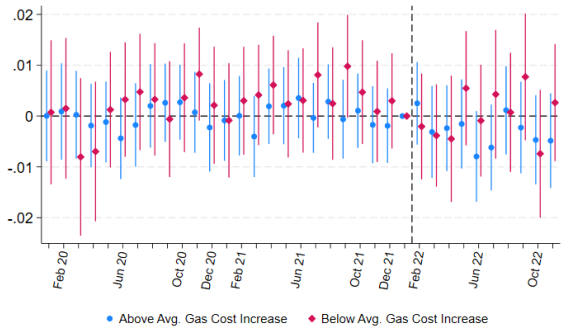
Figure A21: Impact of Heating Costs by above/below Avg. Treatment Intensity



(a) Rental Price



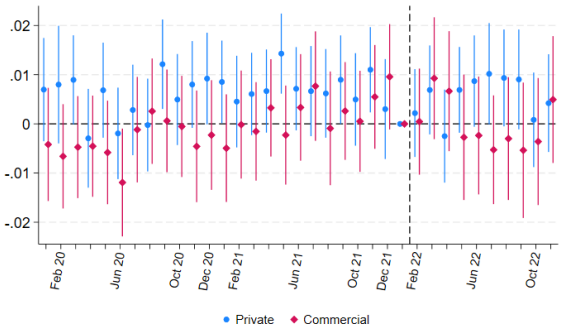
(b) Time on Market



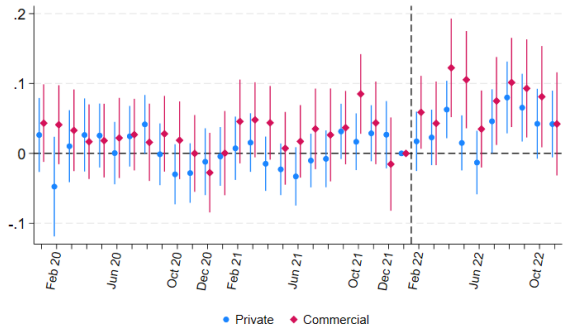
(c) Clicks per day

Notes: The Figures plot the estimates and 95% confidence intervals from equation (5) for below and above average local gas cost increase. Standard errors are clustered at the zip code level. The outcome variables are the log rental price per sqm, the log time on market in days, and the log clicks per day. The number of observations is 310,057 for the below-average sample and 326,460 for the above-average sample.

Figure A22: Impact of Local Heating Costs by landlord type



(a) Rental Price



(b) Time on Market

Notes: The Figures plot the estimates and 95% confidence intervals from equation (5) for private and commercial landlords respectively. Standard errors are clustered at the zip code level. The outcome variables are the log rental price per sqm, the log time on market in days, and the log clicks per day.